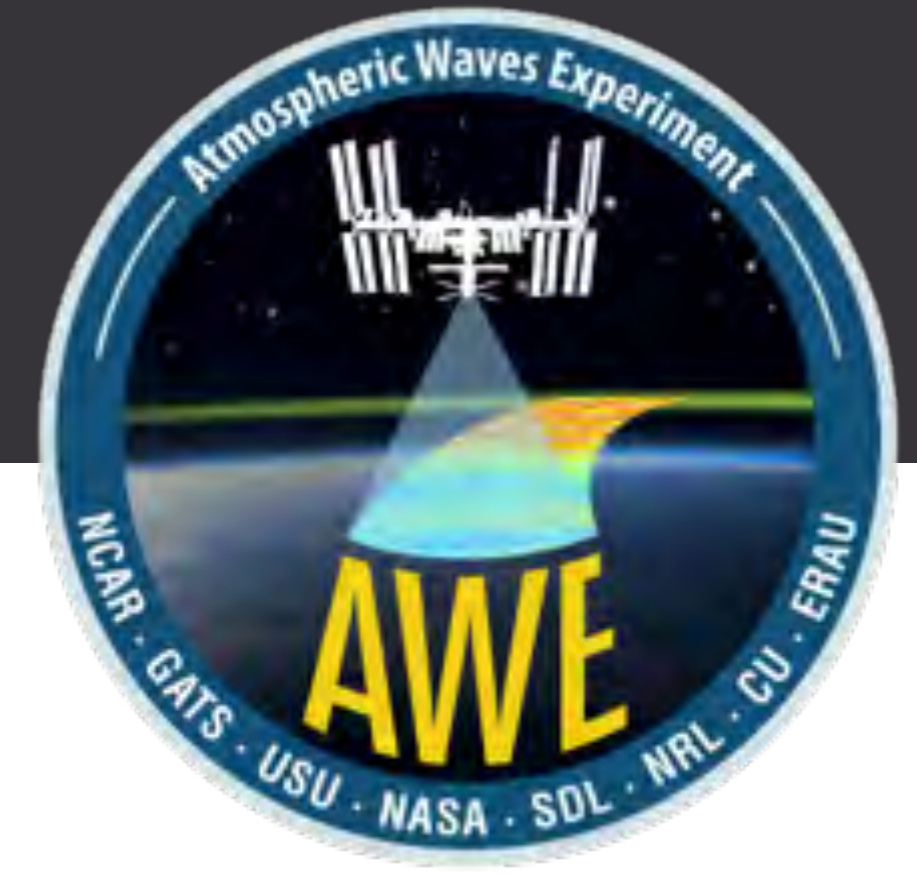




Reconstruction of Atmospheric Gravity Waves from Deep Convection Over CONUS Using **AWE**, TEC, and AIRS Data

CEDAR 2025 June 22–27, 2025 | Des Moines, IA

Jaime Aguilar Guerrero¹, Björn Bergsson¹, Sehin Mesfin¹, Pavel Inchin², Matthew Zettergren¹, Ludger Scherliess³, Yucheng Zhao³, Dominique Pautet³

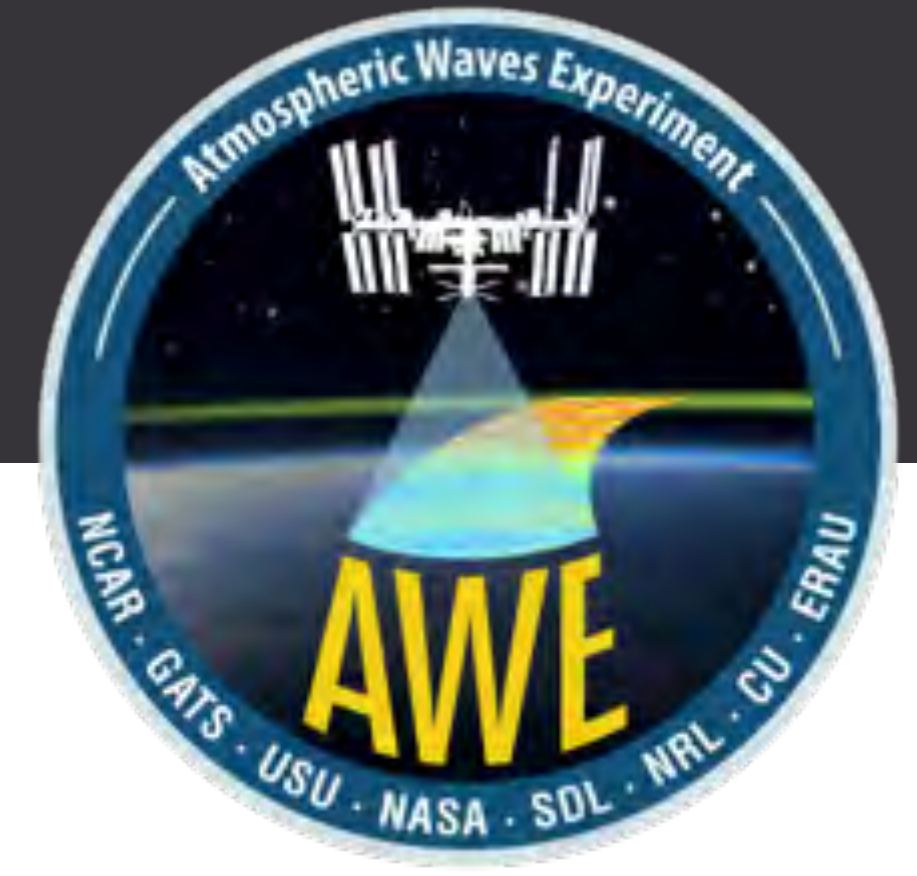
¹*Embry-Riddle Aeronautical University, Daytona Beach, Florida, USA*

²*Computational Physics, Inc., Springfield, Virginia, USA*

³*Utah State University, Logan, Utah, USA*



The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.



Multi-instrument Imaging of Deep Convective Events and Their Gravity Wave Responses Over CONUS Using **AWE**, AIRS, TEC, and LLITED mission

CEDAR 2025 June 22–27, 2025 | Des Moines, IA

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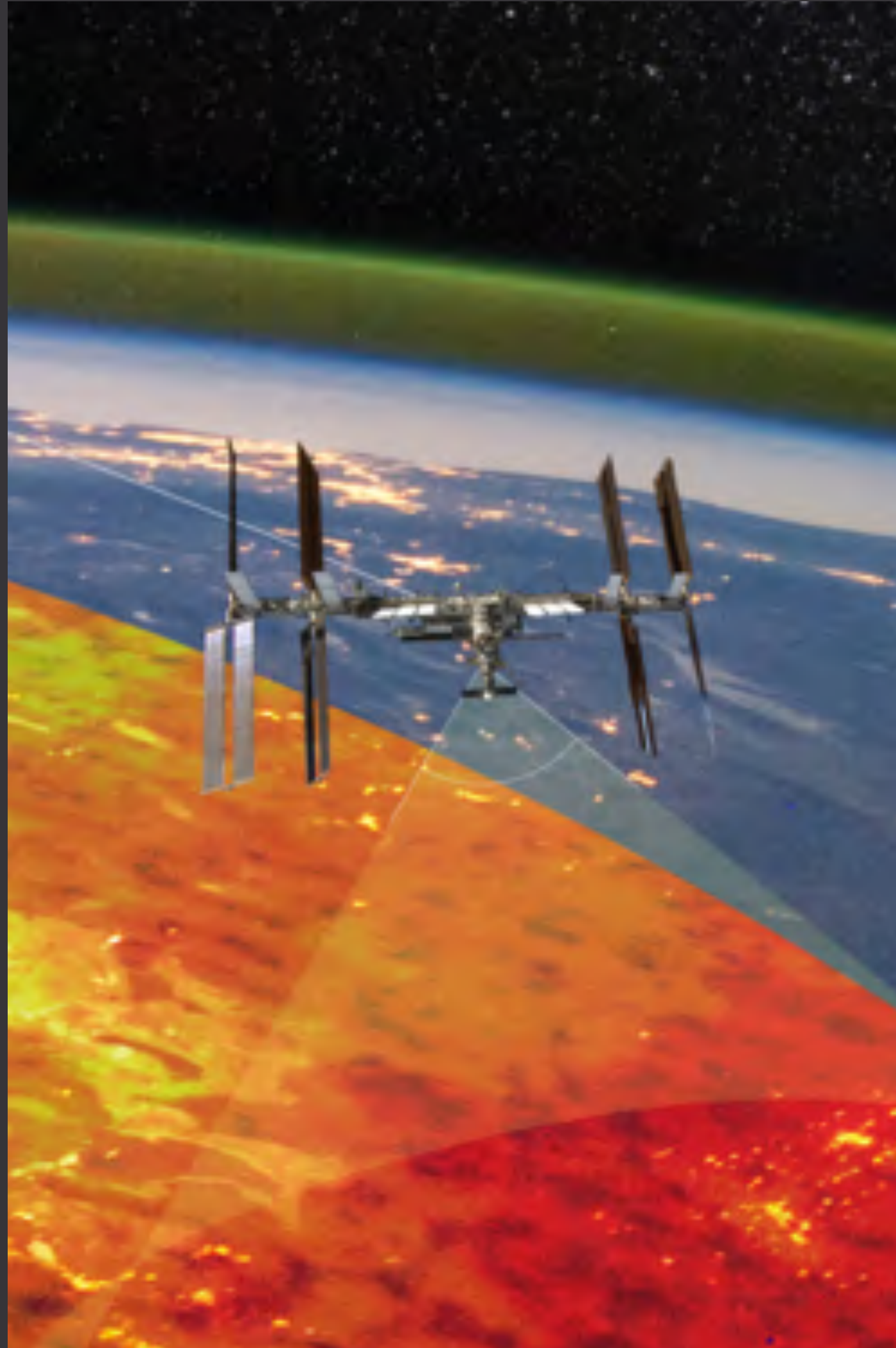
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AWE science: multi-instrument case study

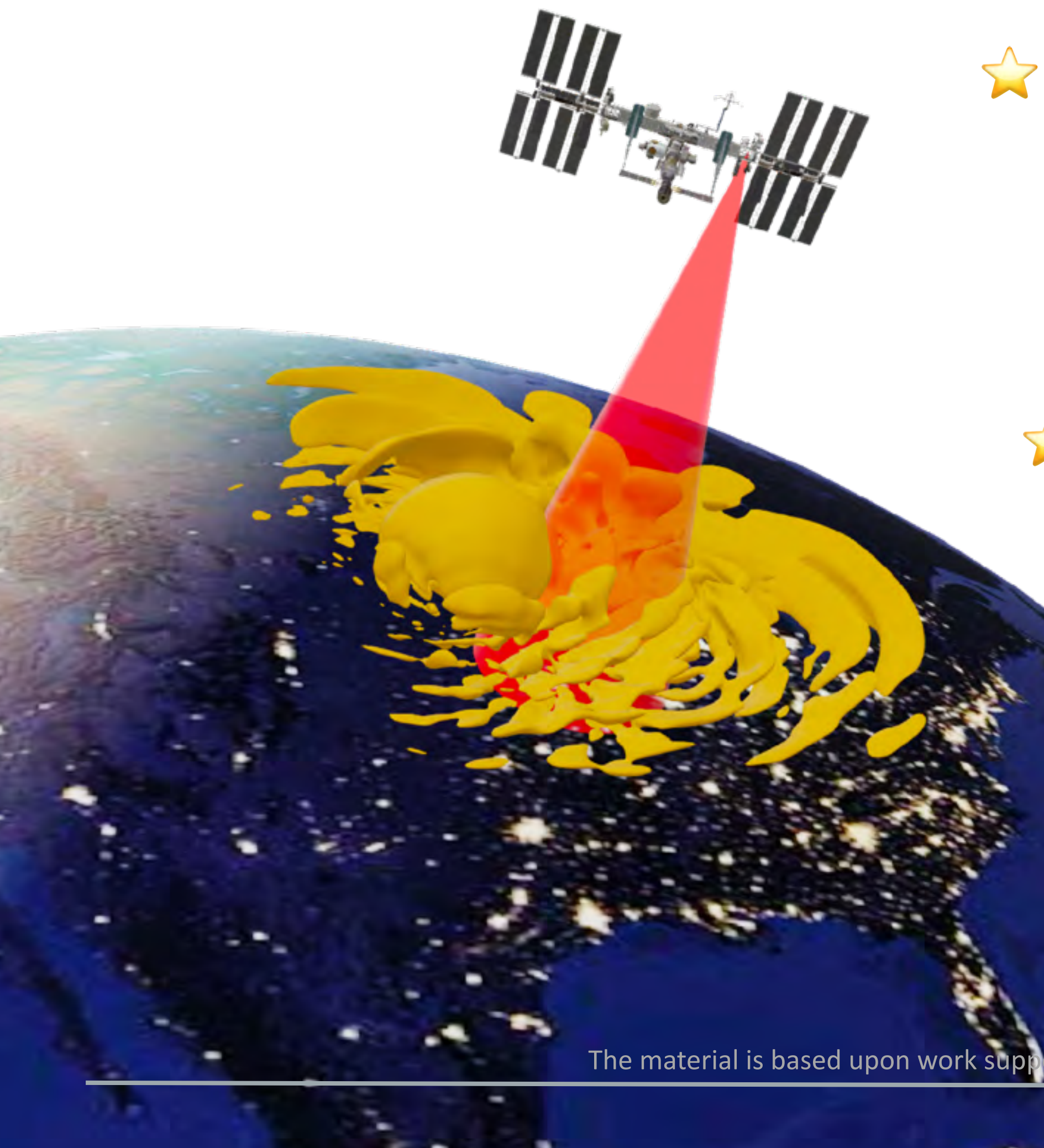
Presentation outline:

This presentation is divided into four main topics:

- **Background:** AWE & tropospheric sources.
- **Observations:** 05/26/24, AWE, AIRS, GNSS & LLITED.
- **Methods:** time-distance diagrams and spectral analysis.
- **Results:** gravity wave parameters & reconstruction.

AWE Atmospheric Waves Experiment

Observing how atmospheric waves from Earth's weather systems influence the dynamics and variability of the ionosphere

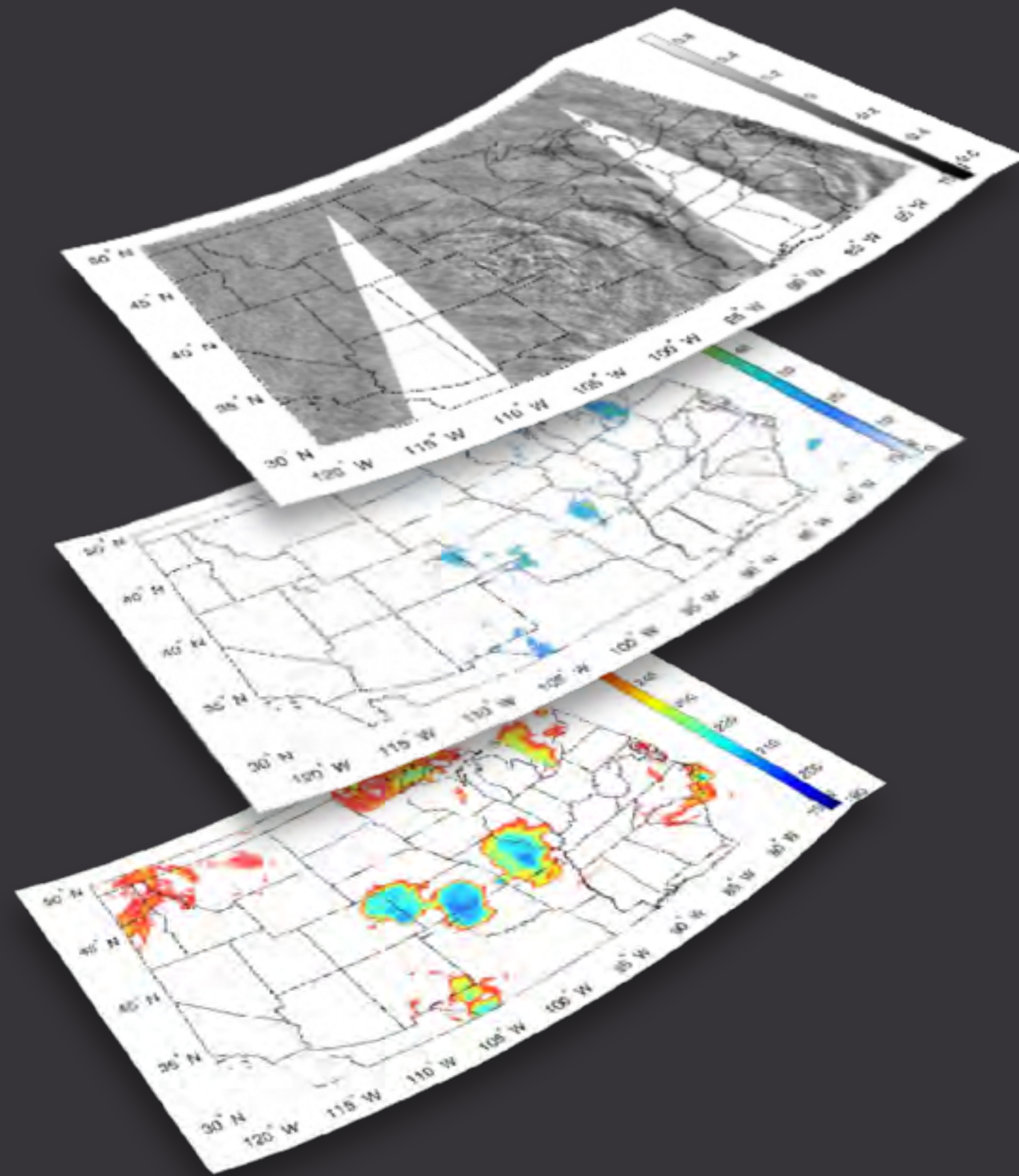


- ★ **Objectives:** 1. Quantify the seasonal and regional variabilities and influences of GWs near the mesopause. 2. Identify the dominant dynamical processes controlling GWs. Estimate the broader role of GWs in the ITM.
- ★ **Instrument:** Advanced Mesospheric Temperature Mapper (AMTM): nighttime OH airglow, 600 km FOV, 2×2 km resolution.
- ★ **ISS orbit:** ~15 orbits over ~90 min period; $\pm 55^\circ$ latitude.
- ★ **Mission Profile:** two years; launched November 9th, 2023.

AWE science: tropospheric sources

Some of the most common sources:

- **Convective** sources: thunderstorms, tornados, hurricanes, typhoons, monsoons, etc..
- **Orographic** waves.
- **Secondary** gravity waves.



(top) CO₂ radiance as measured by AIRS (4.3 μm BWT) showcasing gravity waves. (middle) NEXRAD reflectivity (dBz) as a source for MAGIC. (bottom) Cloud top temperature (AIRS 8.1 μm BT) as a source for MAGIC. *Courtesy of Christopher Heale CEDAR 2022, based on his (2019) GRL paper and the work of Stephan & Alexander (2015).*

The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.



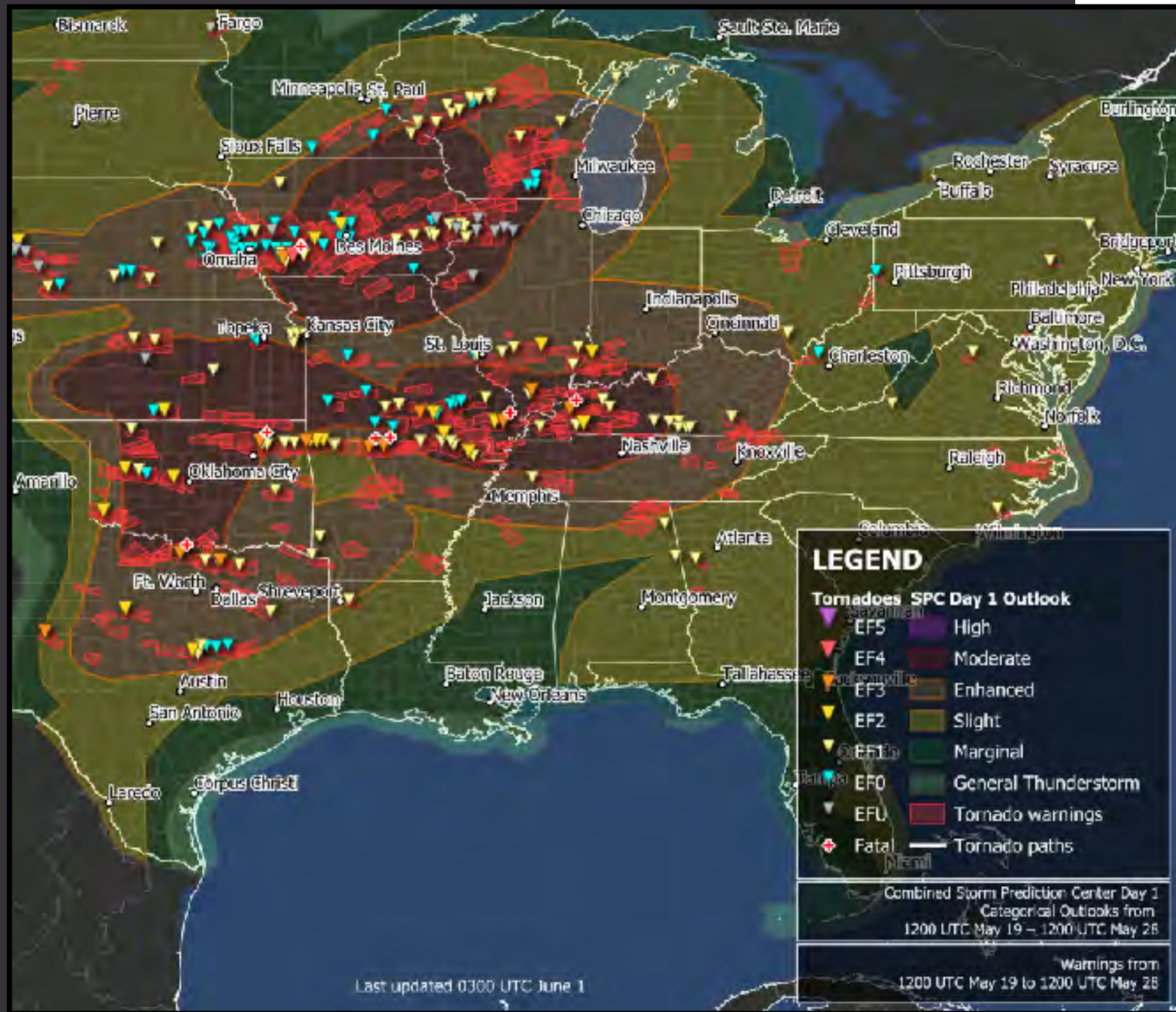
*Model credit: Model for Acoustic-Gravity waves Interactions
Coupling (MAGIC) model by (Snively and Zettergren, 2015)*

07/08/2016 02:09:00 UT

The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

AWE science: case study — May, 2024

A powerful and deadly convective system across the Midwest USA



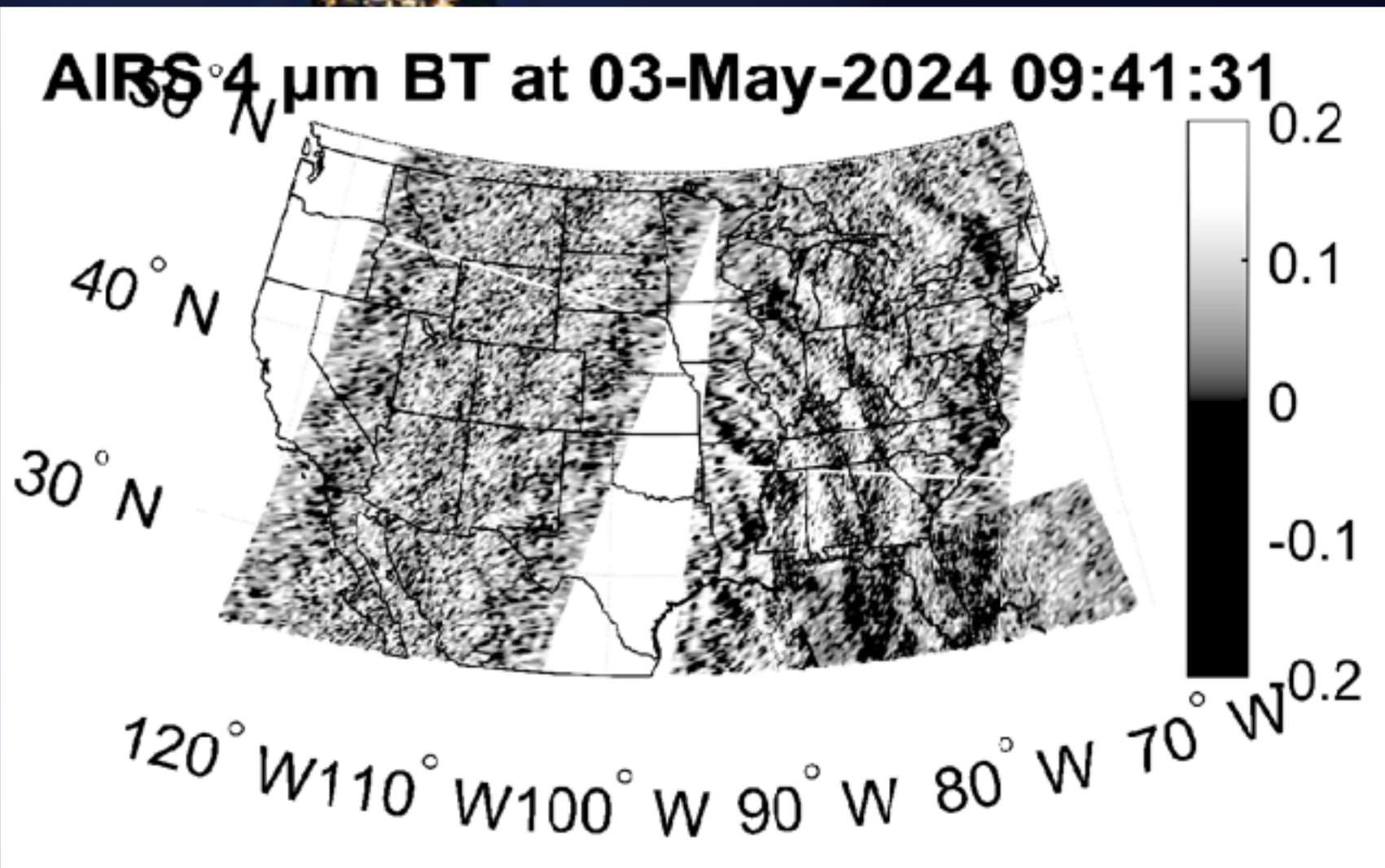
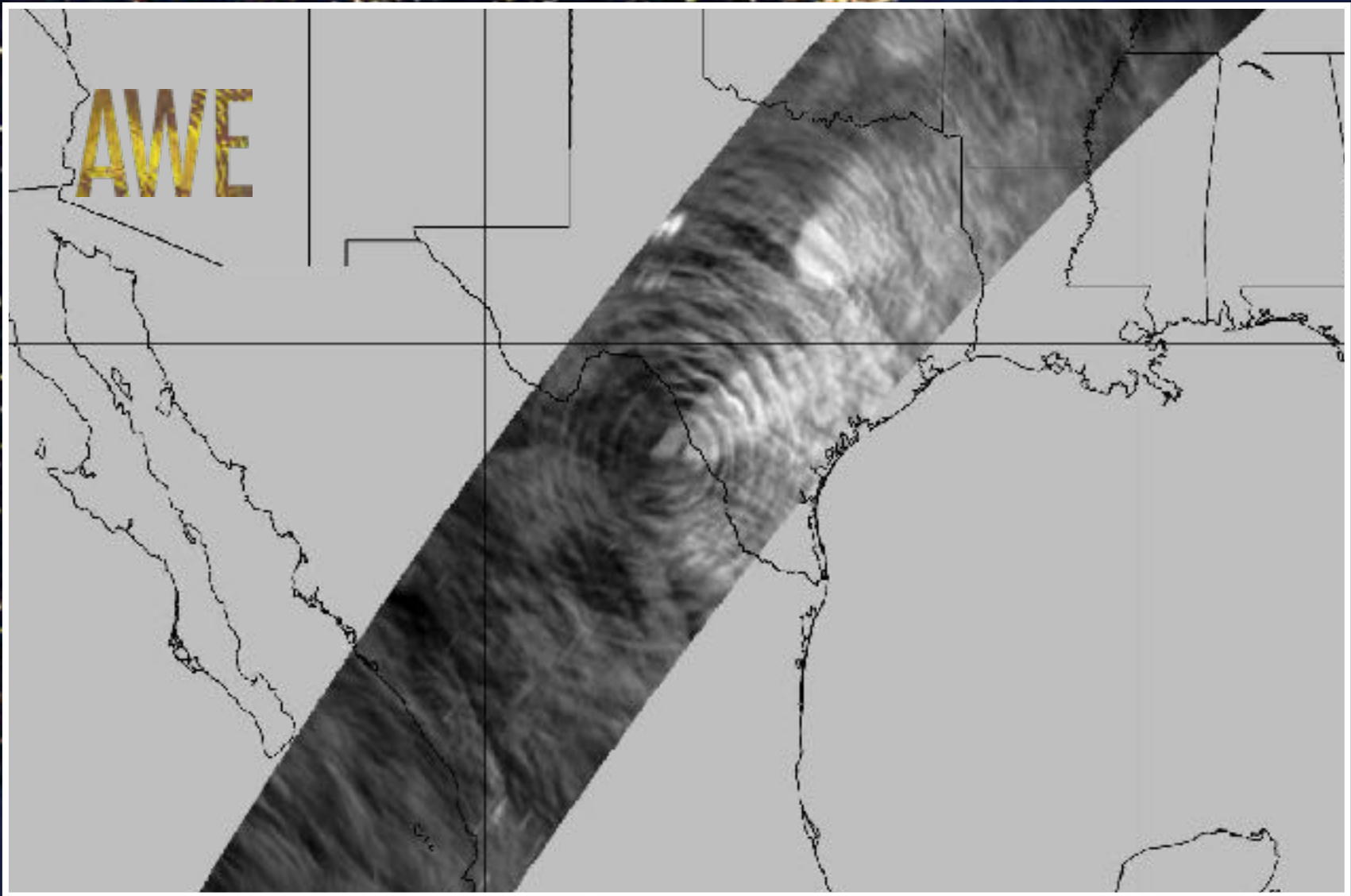
Map of tornado warnings and confirmed tornadoes from the outbreak sequence of May 19th–27th, 2024. By Wxtrackercody - Own work, CC BY 4.0, <https://commons.wikimedia.org/w/index.php?curid=148937773>

The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

- Persistent Gulf moisture plume + powerful spring jet over the Plains/Midwest produced repeated high-instability/strong-shear environments, yielding daily rounds of supercells, large hail and damaging wind lines.
- 534 confirmed **tornadoes** (including 2 EF4)
- 2nd-most active May on record; **3rd-most active month overall**

May 3rd, 2024

AWE

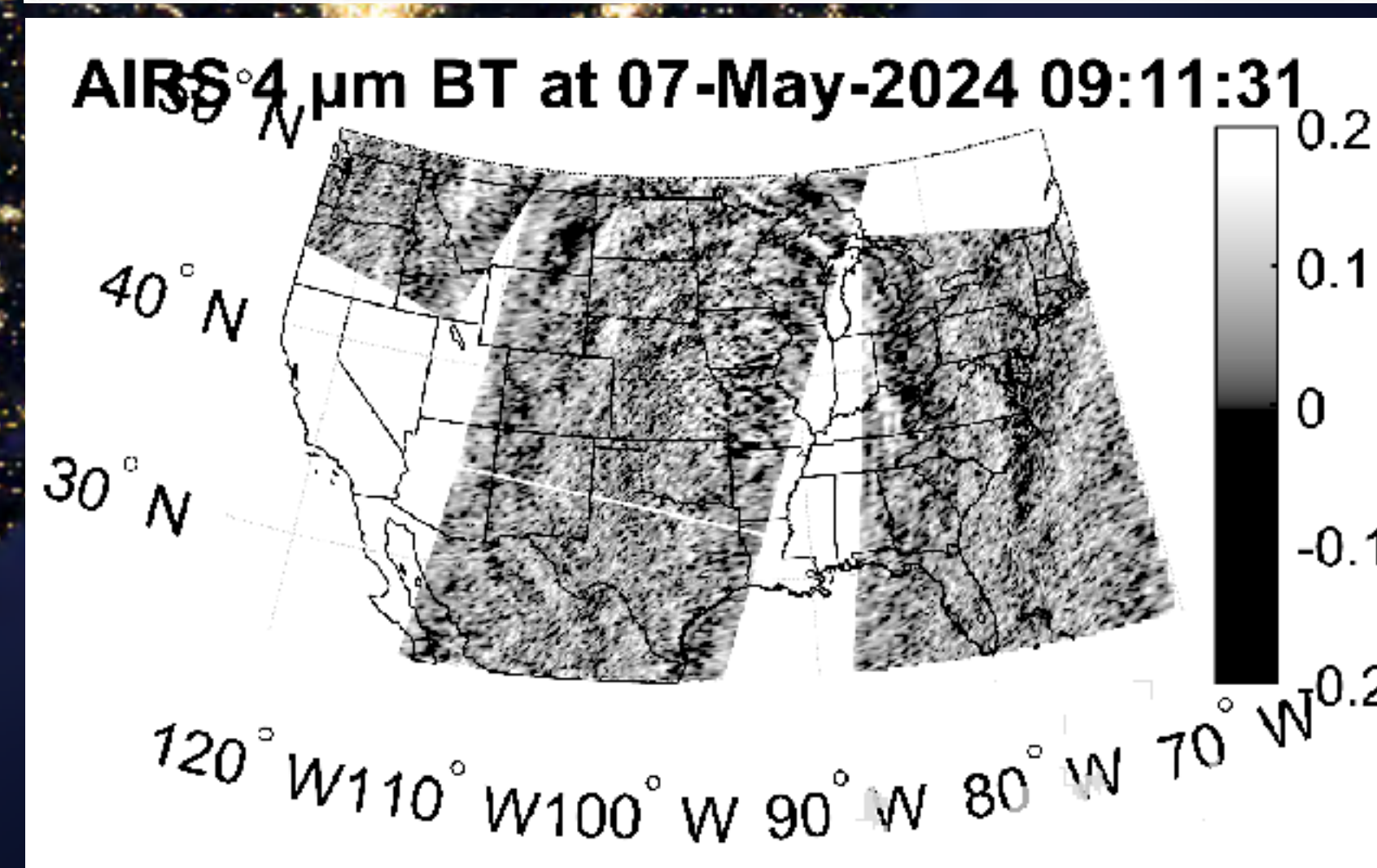
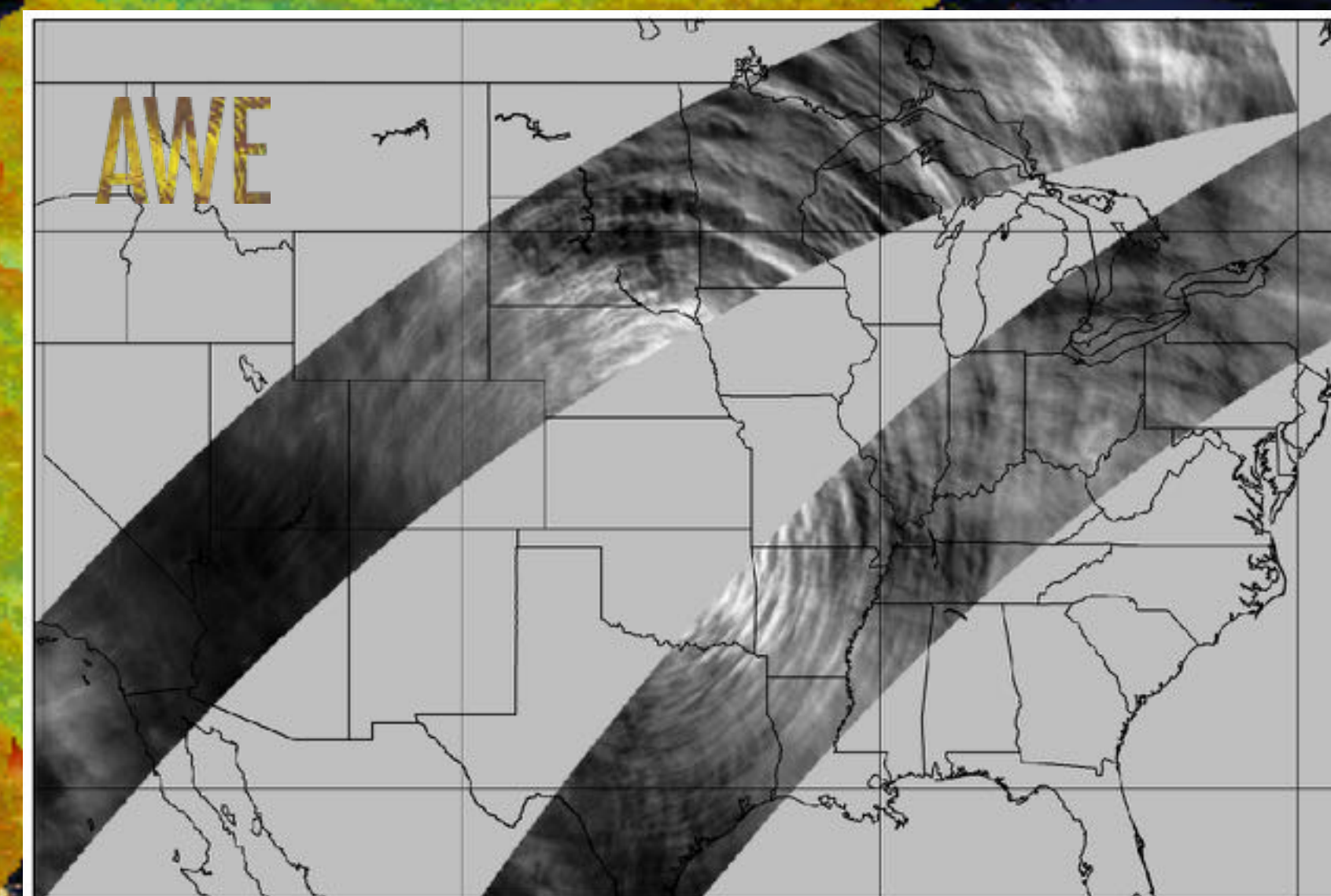


The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

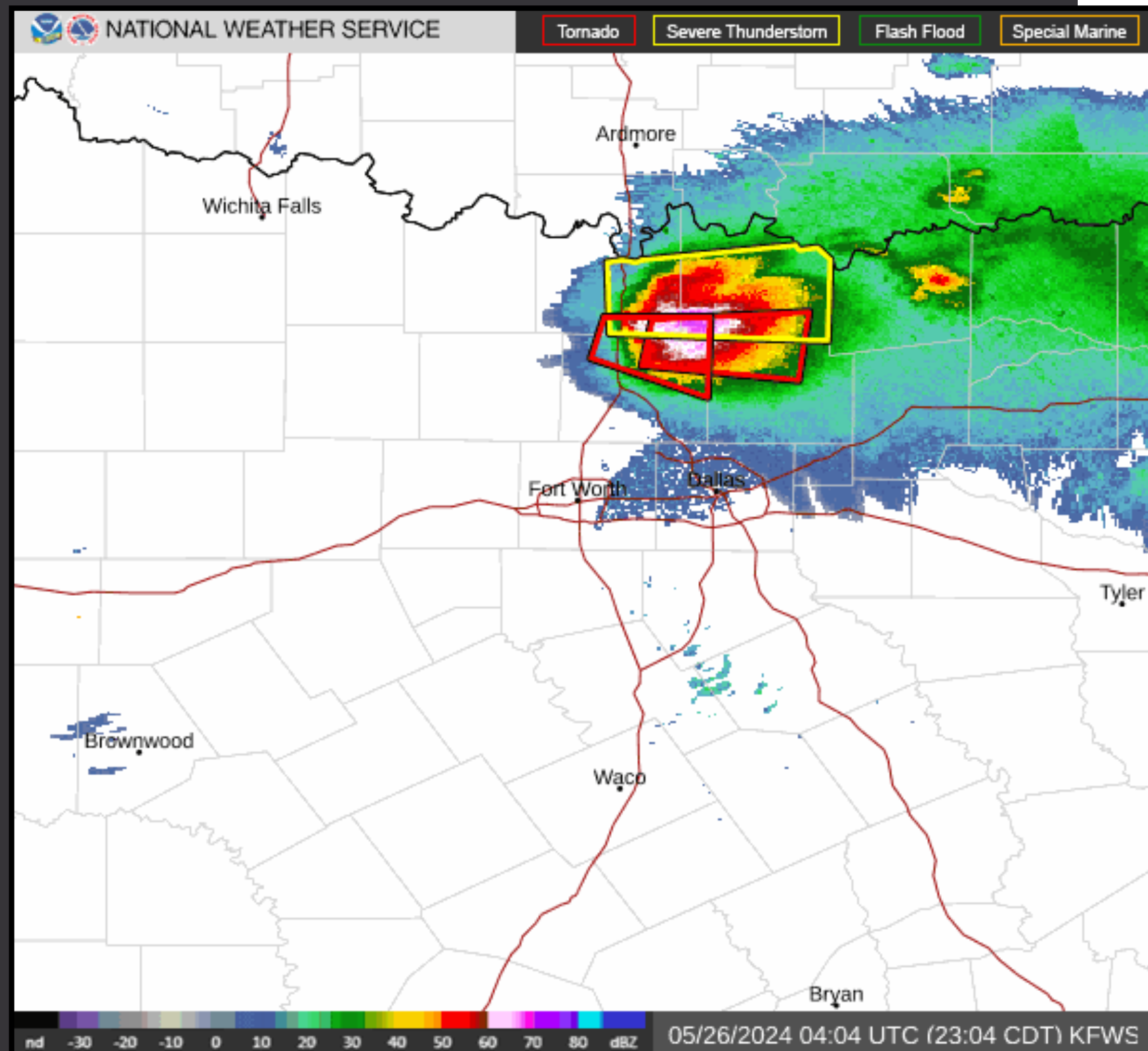
May 7th, 2024

2-Orbits

AWE



The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.



Radar imagery of the tornadic supercell that was producing the EF3 Valley View tornado as it moved eastward after passing south of Valley View.. *By National Weather Service Fort Worth TX - <https://www.weather.gov/fwd/>, Public Domain, <https://commons.wikimedia.org/w/index.php?curid=149912229>*

The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

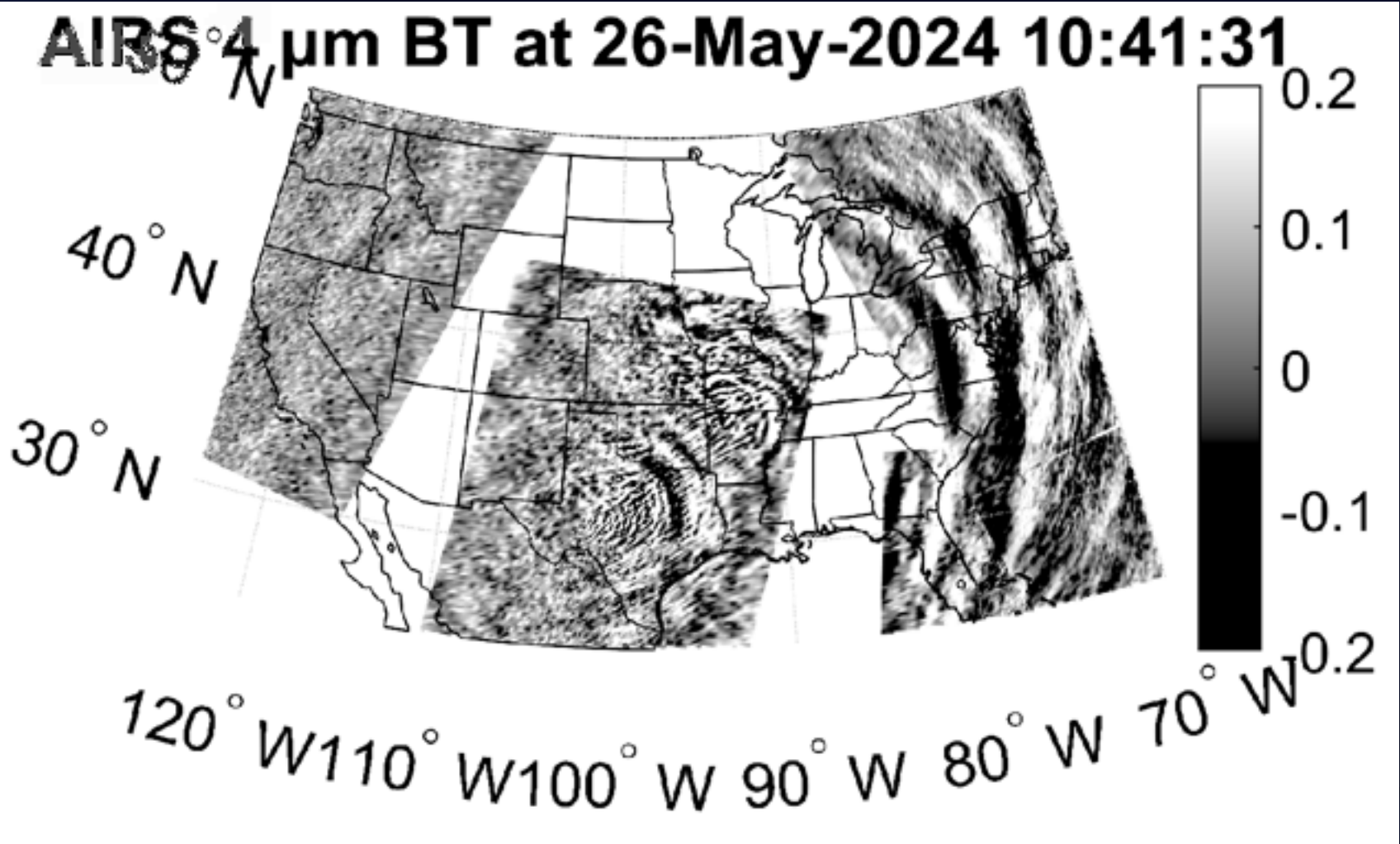
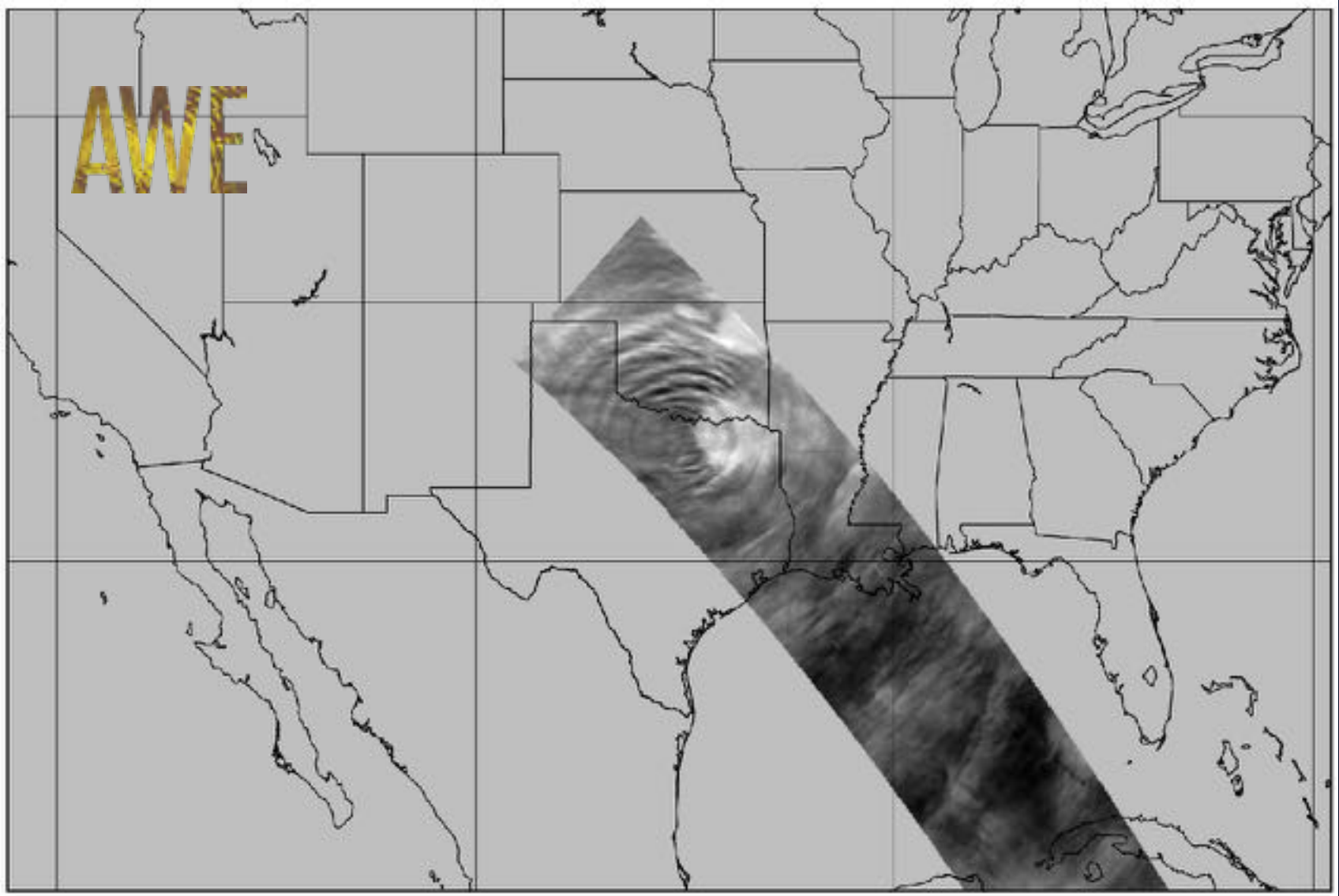
AWE science: case study — May 26th, 2024

EF3 category tornado North of Dallas, TX

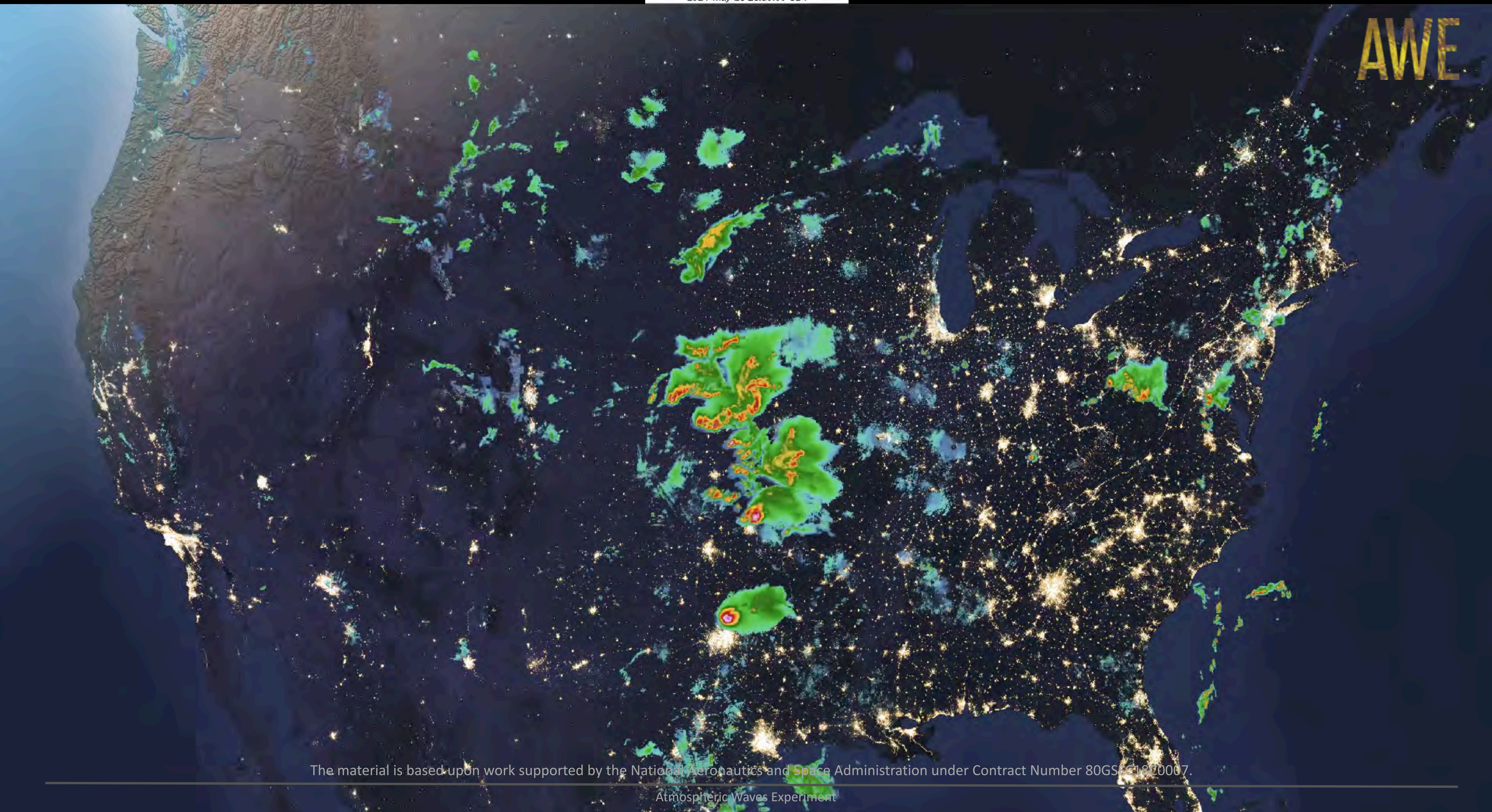
- AWE observed the Valley View/Cooke Co., TX tornado (overnight May 25th– 26th).
- Low-end EF3 (peak 140 mph).
- Struck at **9:39 pm CDT (3:00 UTC)**.
- **1:36 hrs** on ground.
- **40 mi** track.

May 26th, 2024

AWE



The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

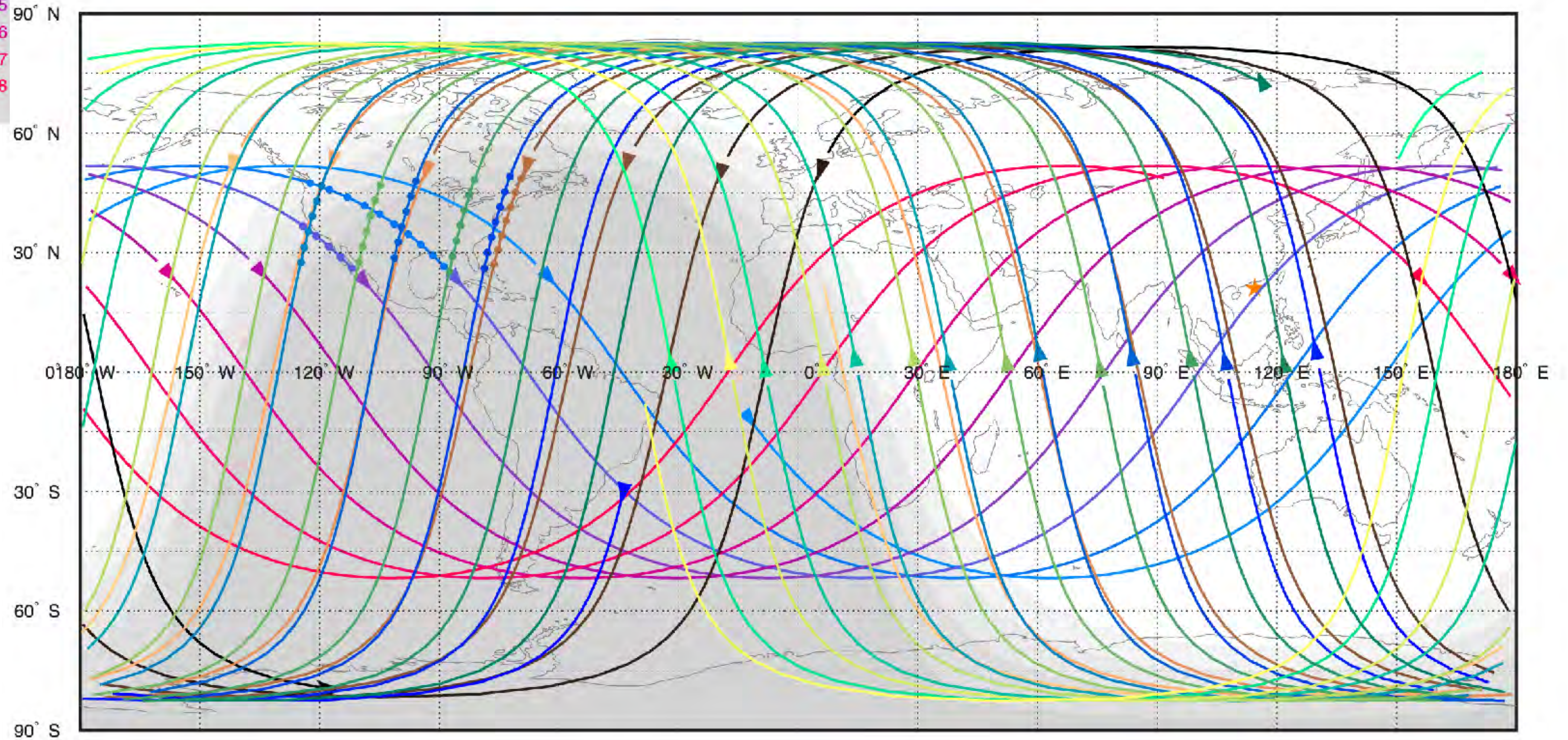




The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

ISS + AQUA (AIRS) + LLITED A&B orbits May 26th, 2024 1:30–13:30 UTC

ISS (ZARYA) - Orbit 45512
ISS (ZARYA) - Orbit 45513
ISS (ZARYA) - Orbit 45514
ISS (ZARYA) - Orbit 45515
ISS (ZARYA) - Orbit 45516
ISS (ZARYA) - Orbit 45517
ISS (ZARYA) - Orbit 45518
AQUA - Orbit 17348
AQUA - Orbit 17349
AQUA - Orbit 17350
AQUA - Orbit 17351
AQUA - Orbit 17352
AQUA - Orbit 17353
AQUA - Orbit 17354
AQUA - Orbit 17355
LLITED A - Orbit 6289
LLITED A - Orbit 6290
LLITED A - Orbit 6291
LLITED A - Orbit 6292
LLITED A - Orbit 6293
LLITED A - Orbit 6294
LLITED A - Orbit 6295
LLITED A - Orbit 6296
LLITED A - Orbit 6297
LLITED B - Orbit 6290
LLITED B - Orbit 6291
LLITED B - Orbit 6292
LLITED B - Orbit 6293
LLITED B - Orbit 6294
LLITED B - Orbit 6295
LLITED B - Orbit 6296
LLITED B - Orbit 6297
2024-05-26 12:00:00

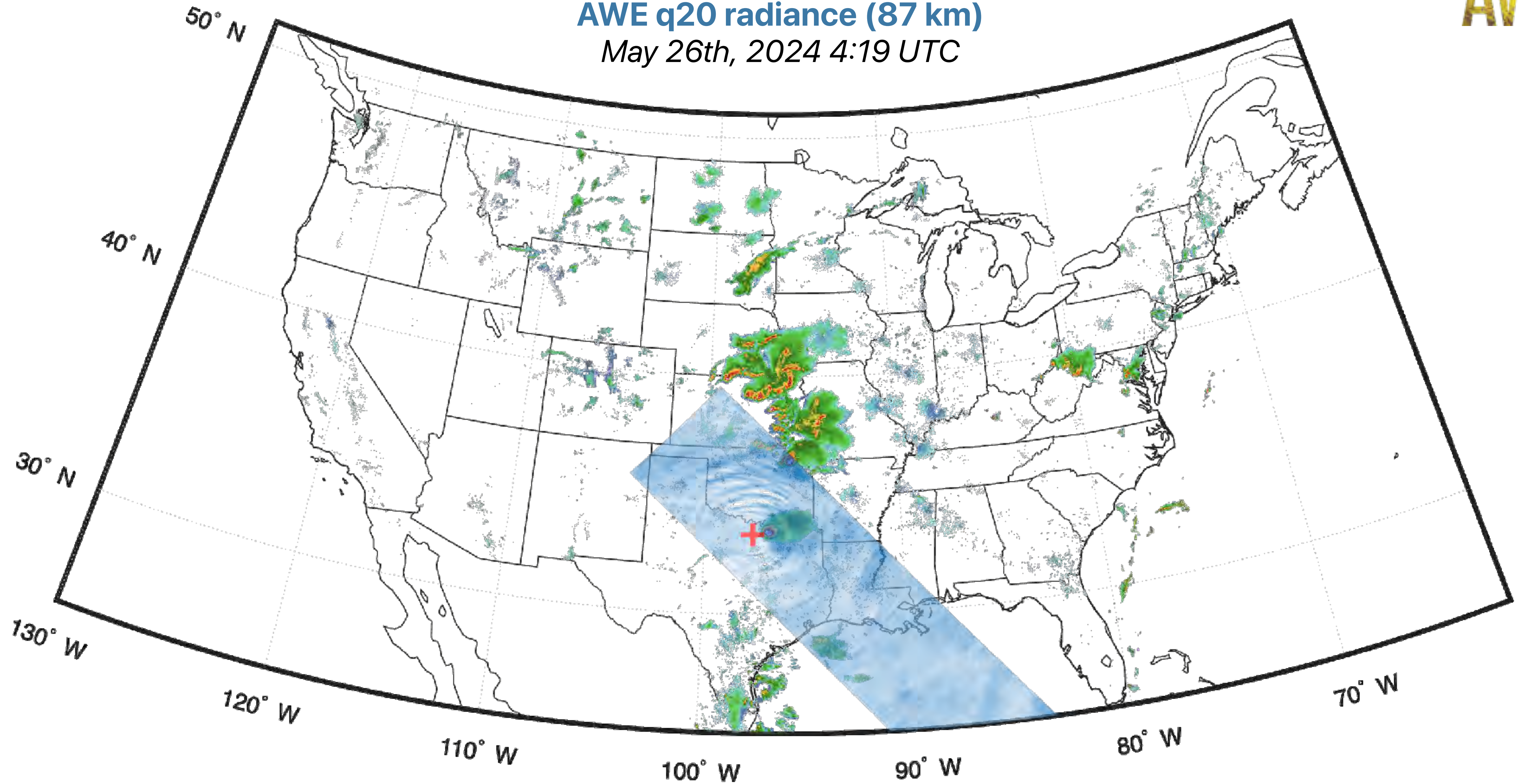


Day/Night terminator at:
2024-05-25 04:19:00 UTC

← (terminator at time of AWE overpass)

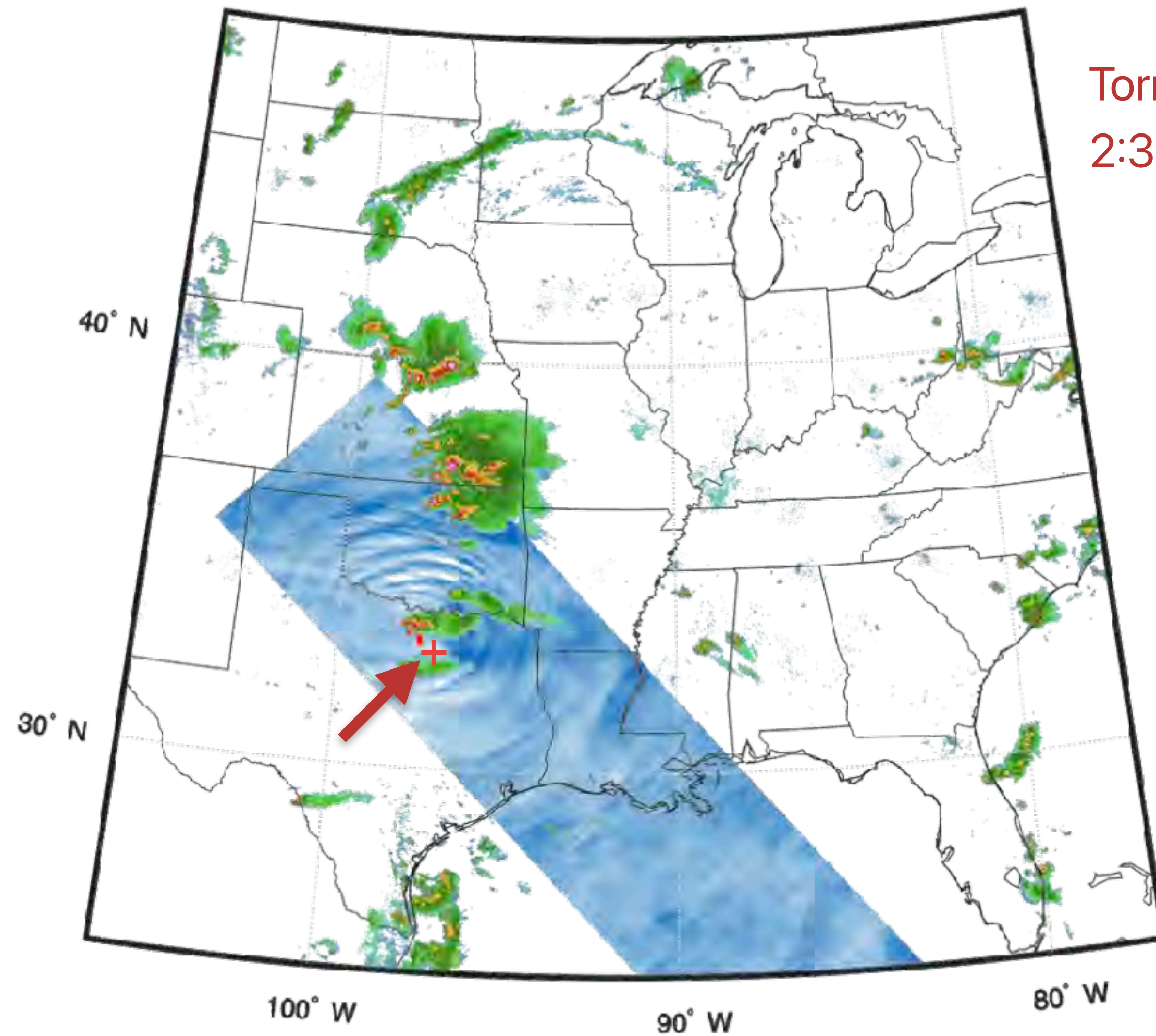
AWE q20 radiance (87 km)

May 26th, 2024 4:19 UTC



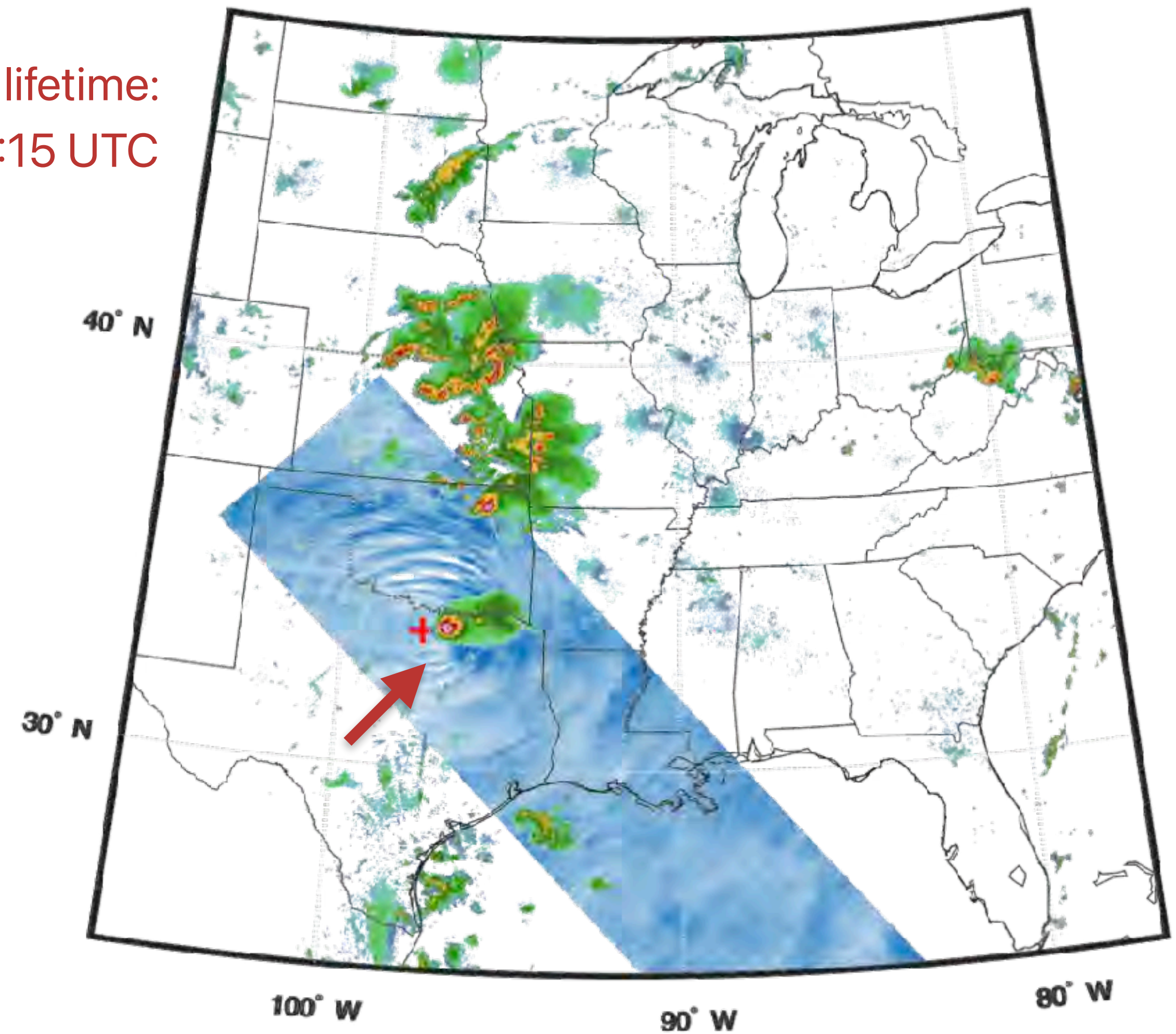
The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

Center of OH gravity wave matches location of tornado at **-2 hrs AWE pass**



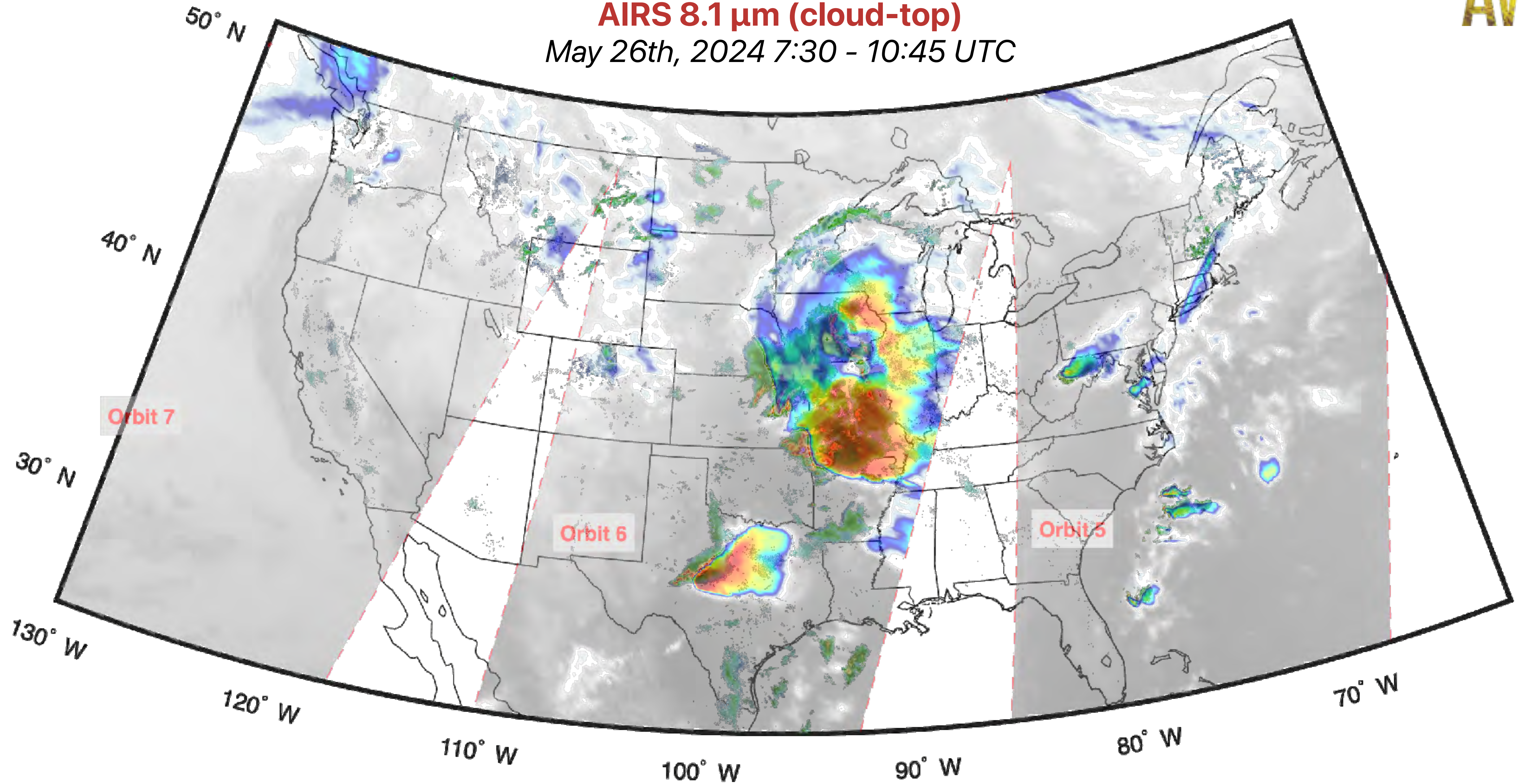
May 26th, 2024 2:10 UTC

Tornado lifetime:
2:39 - 4:15 UTC



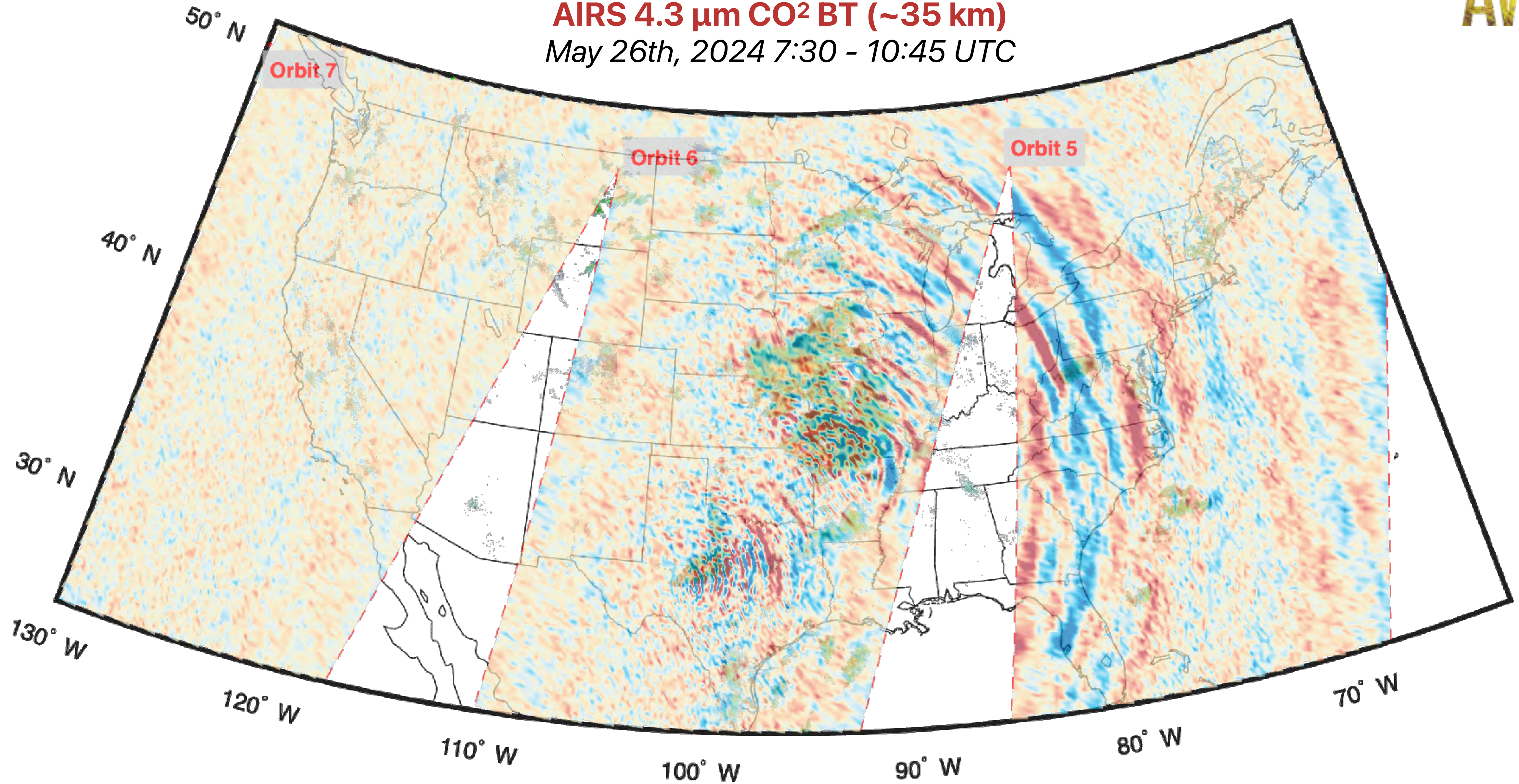
May 26th, 2024 4:19 UTC

AIRS 8.1 μm (cloud-top)
May 26th, 2024 7:30 - 10:45 UTC



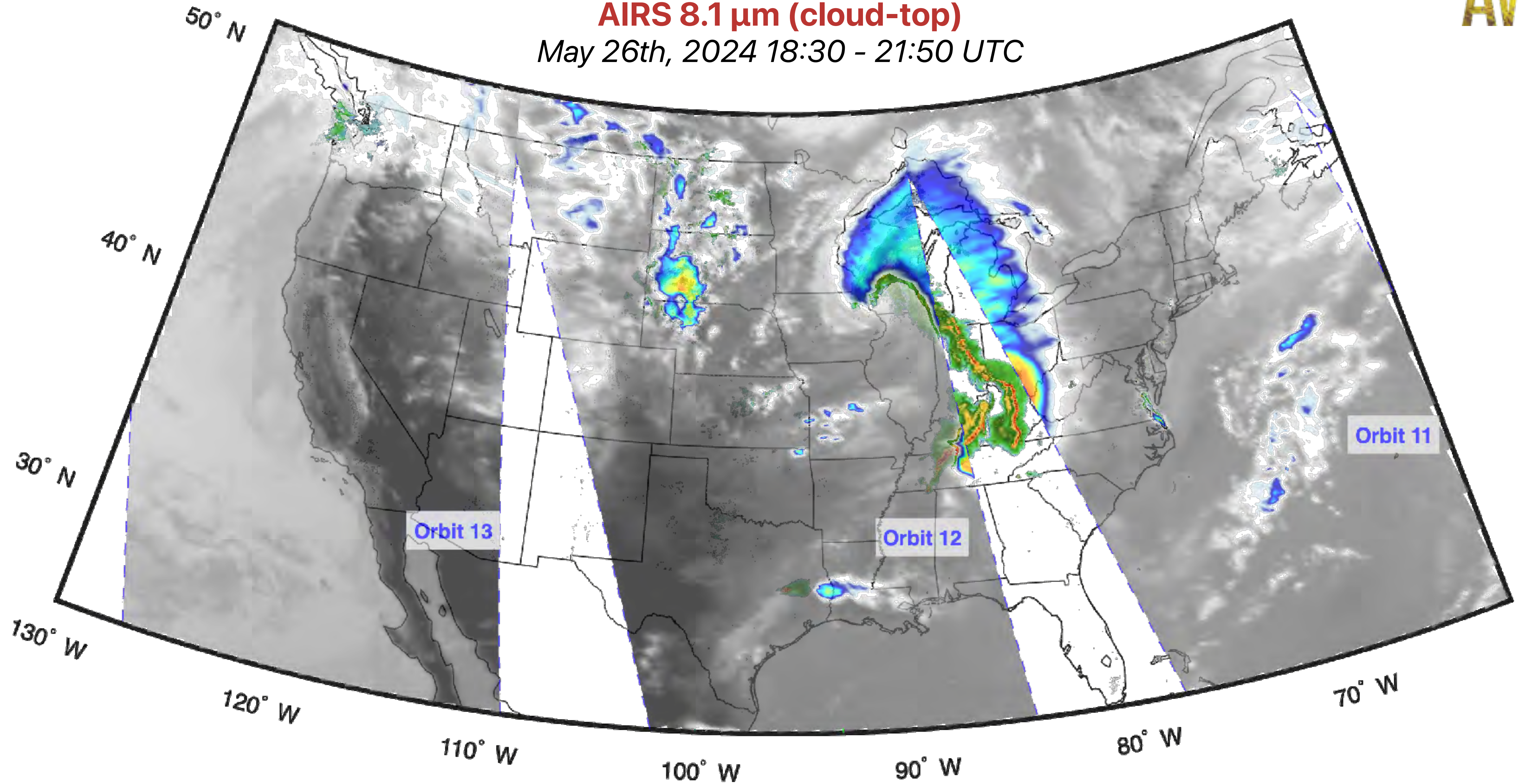
The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

AIRS 4.3 μm CO₂ BT (~35 km)
May 26th, 2024 7:30 - 10:45 UTC



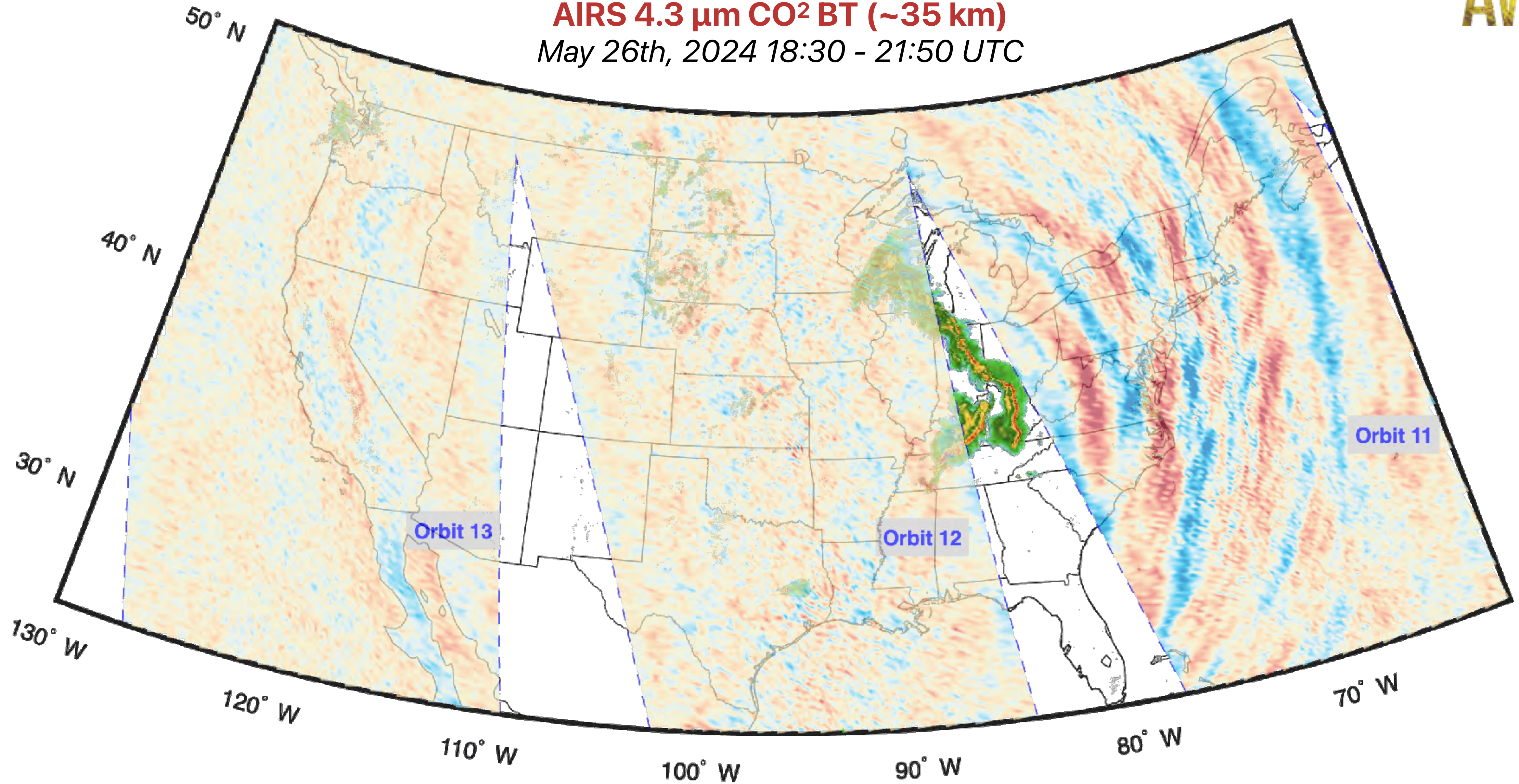
The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

AIRS 8.1 μm (cloud-top)
May 26th, 2024 18:30 - 21:50 UTC



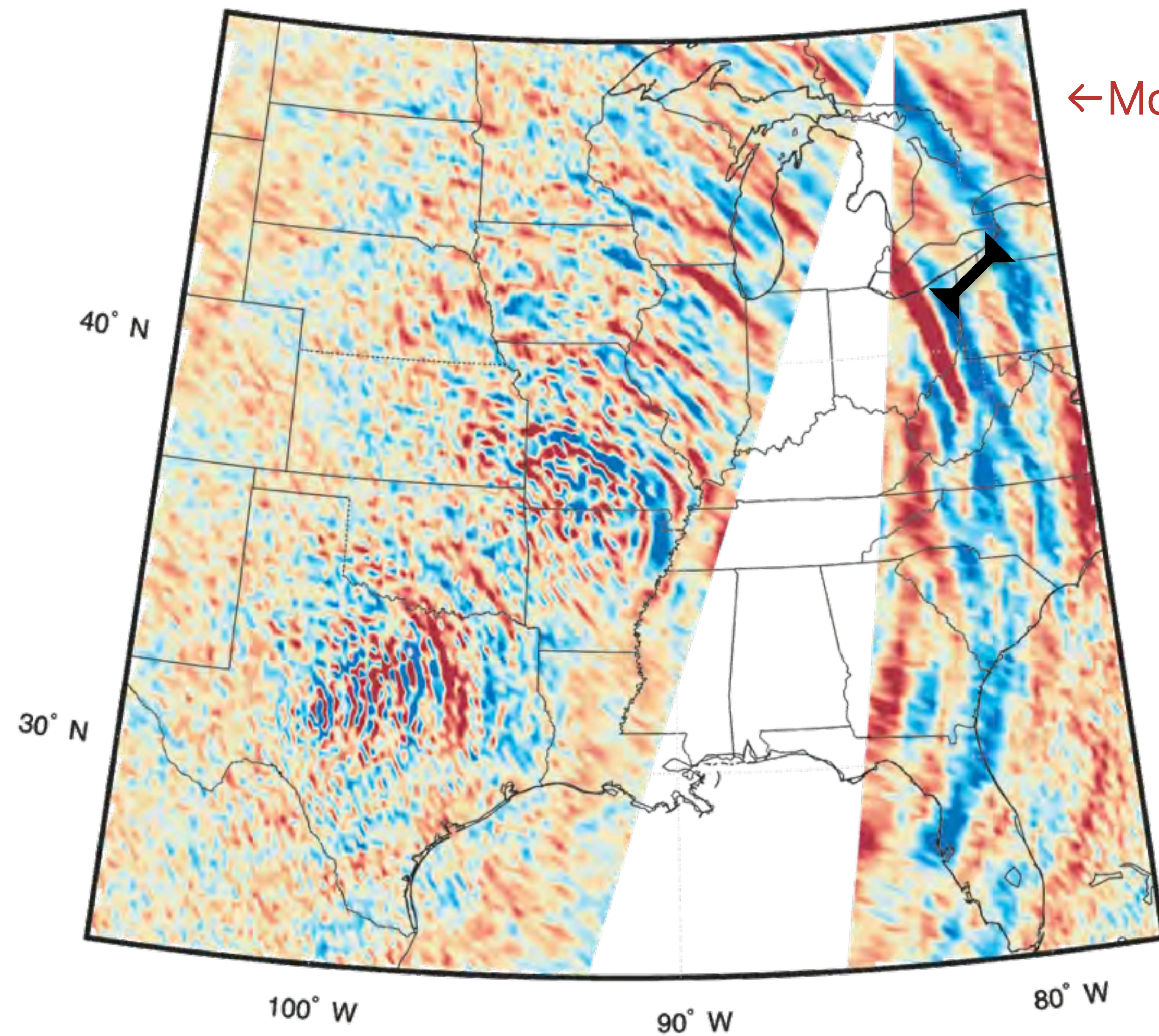
The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

AIRS 4.3 μm CO₂ BT (~35 km)
May 26th, 2024 18:30 - 21:50 UTC



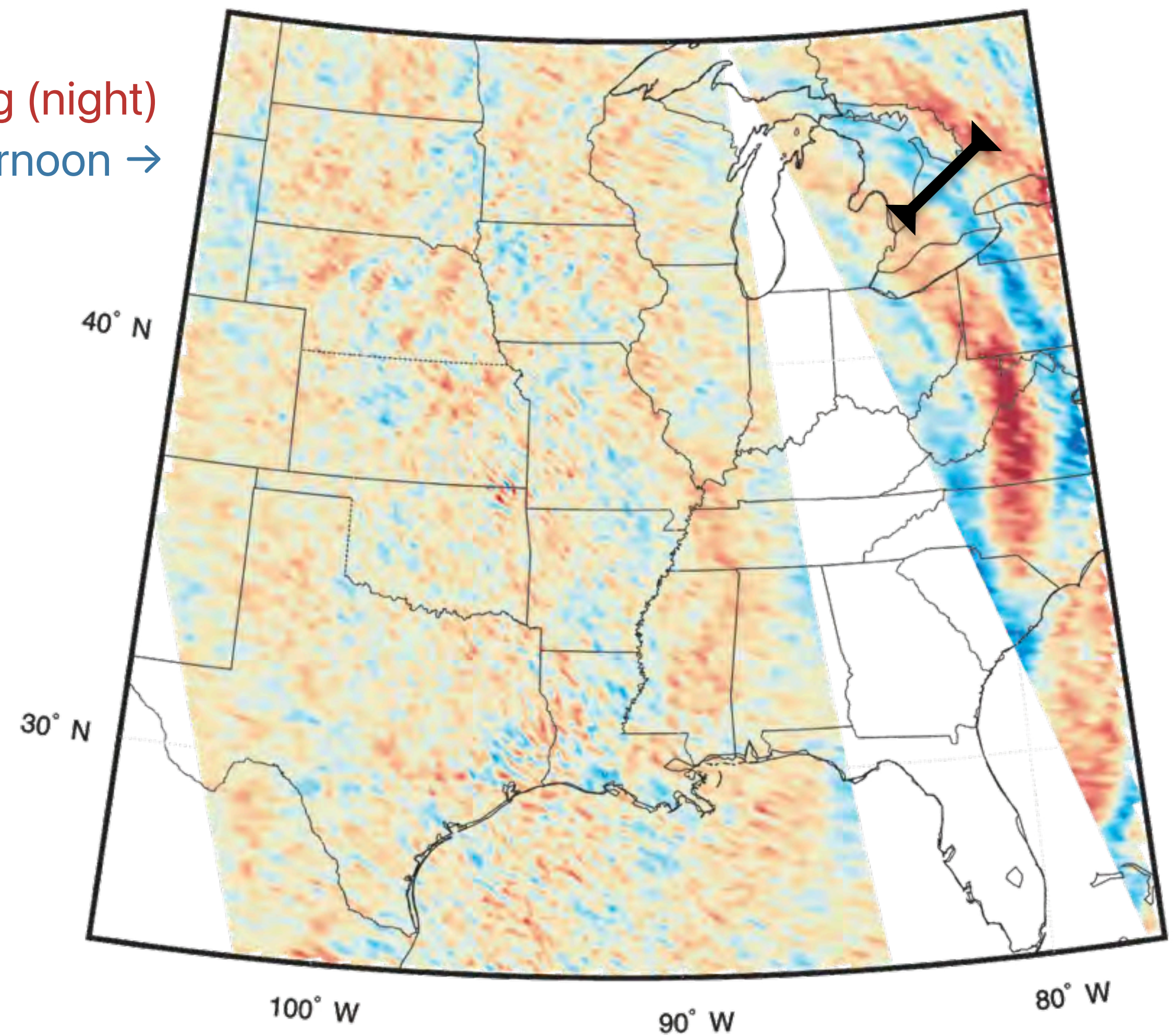
The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

AIRS 4.3 μm measured periods stretch later in the day

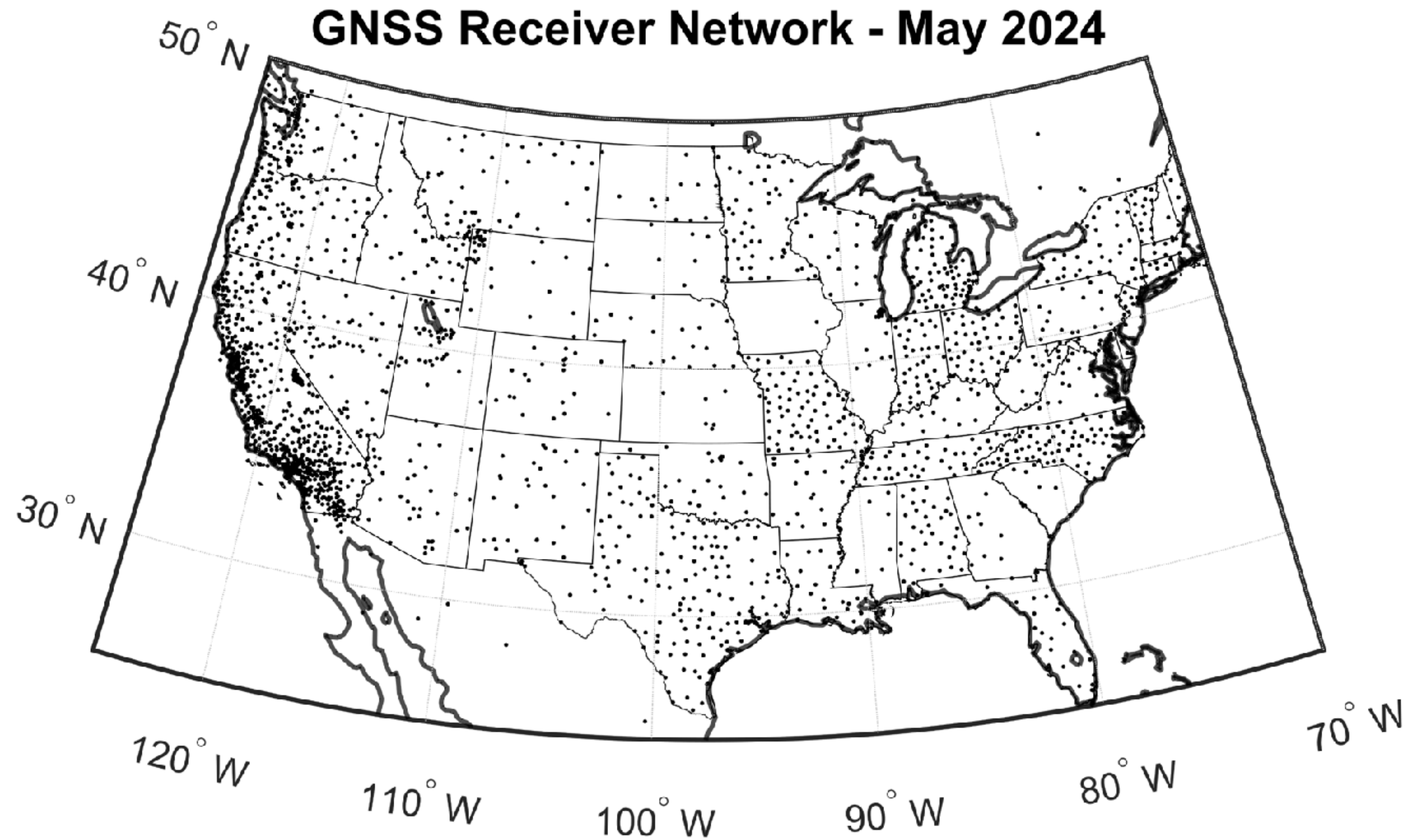


May 26th, 2024 7:30 – 9:10 UTC

← Morning (night)
Afternoon →

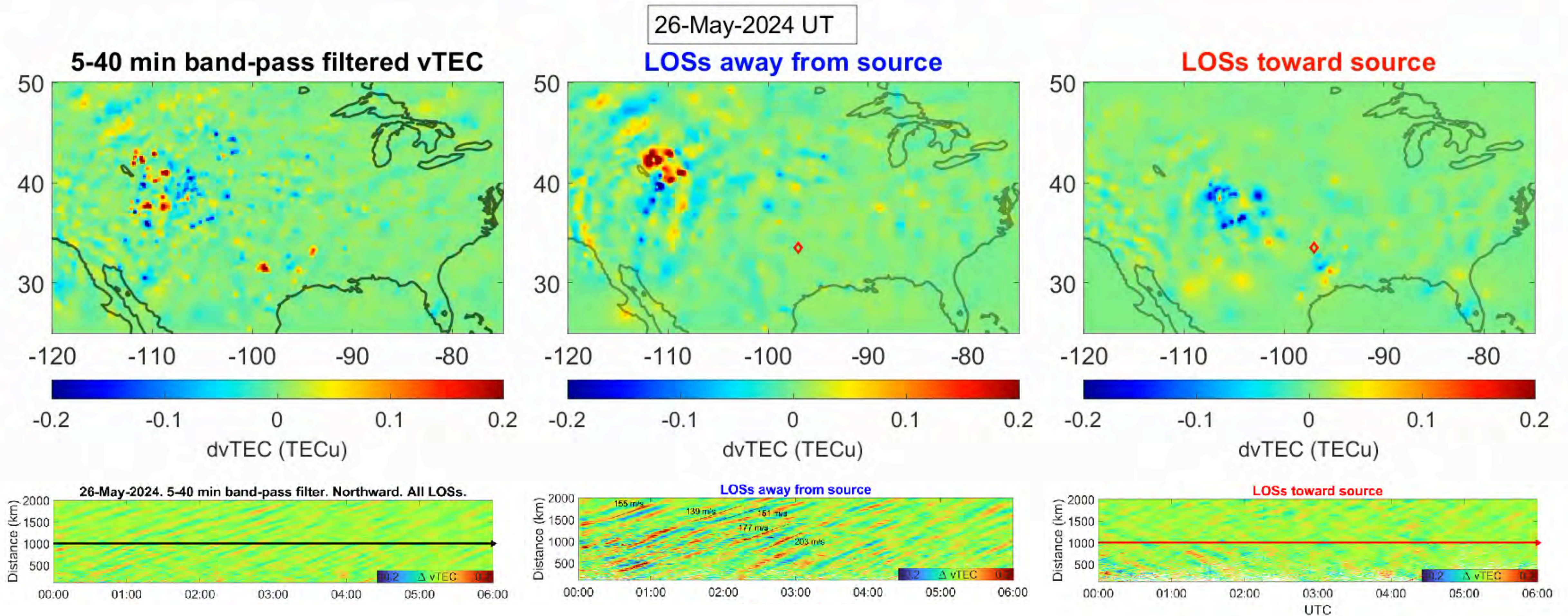


May 26th, 2024 18:30 – 20:10 UTC



The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

GNSS* over CONUS, May 26th 2024

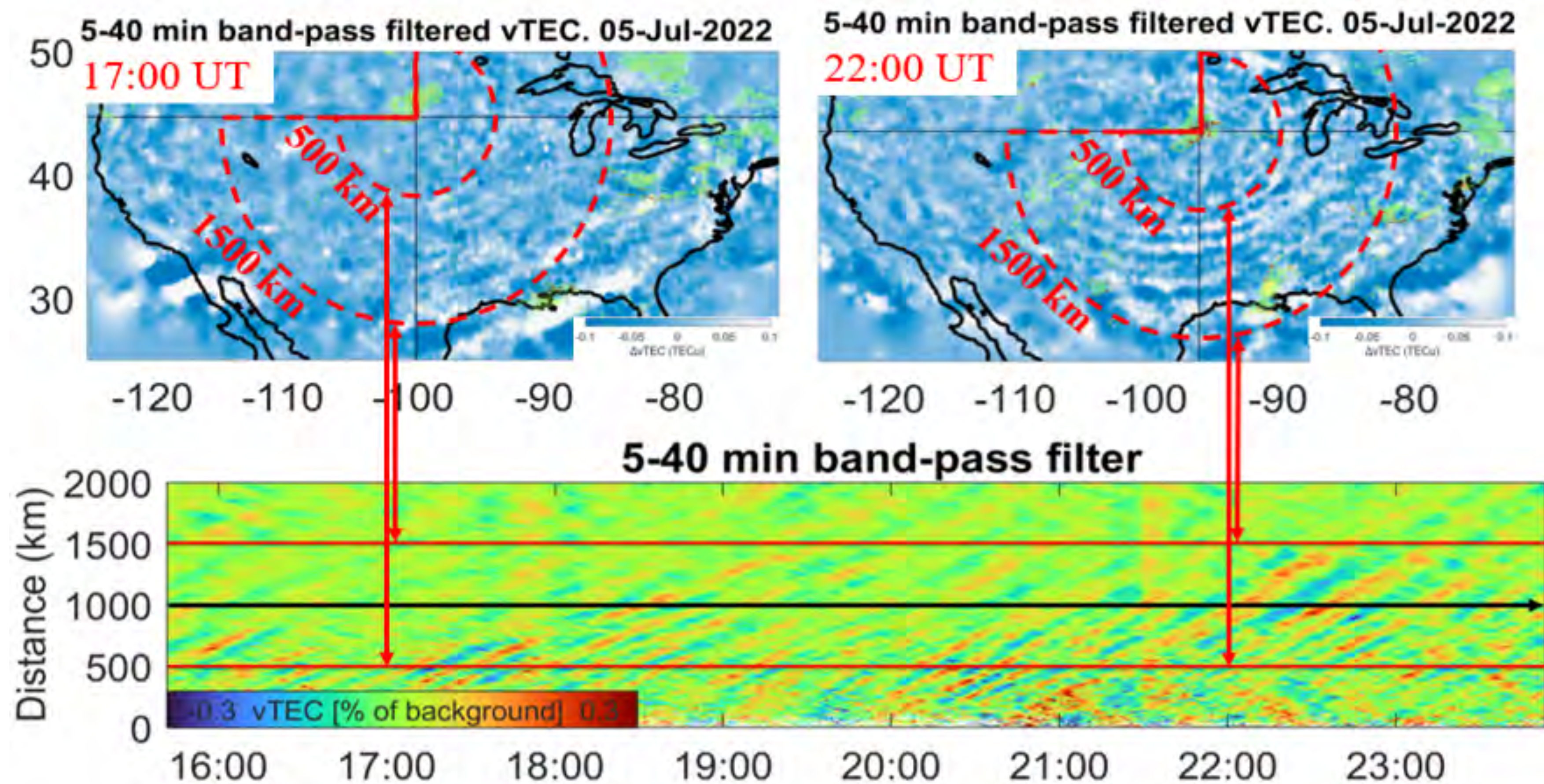


*Inchin, P., Snively, J., & Zettergren, M. (2024). System for Rapid Analysis of Ionospheric Dynamics based on GNSS TEC Signals (S-RAID) [Data set]. Embry-Riddle Aeronautical University Scholarly Commons. <https://doi.org/10.17632/jbx98yscmd.1>

The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

Methods: Time-distance diagram

- A time-distance diagram is constructed from vTEC perturbations by finding the mean value of a concentric region around the convective source location for all distances at all times.

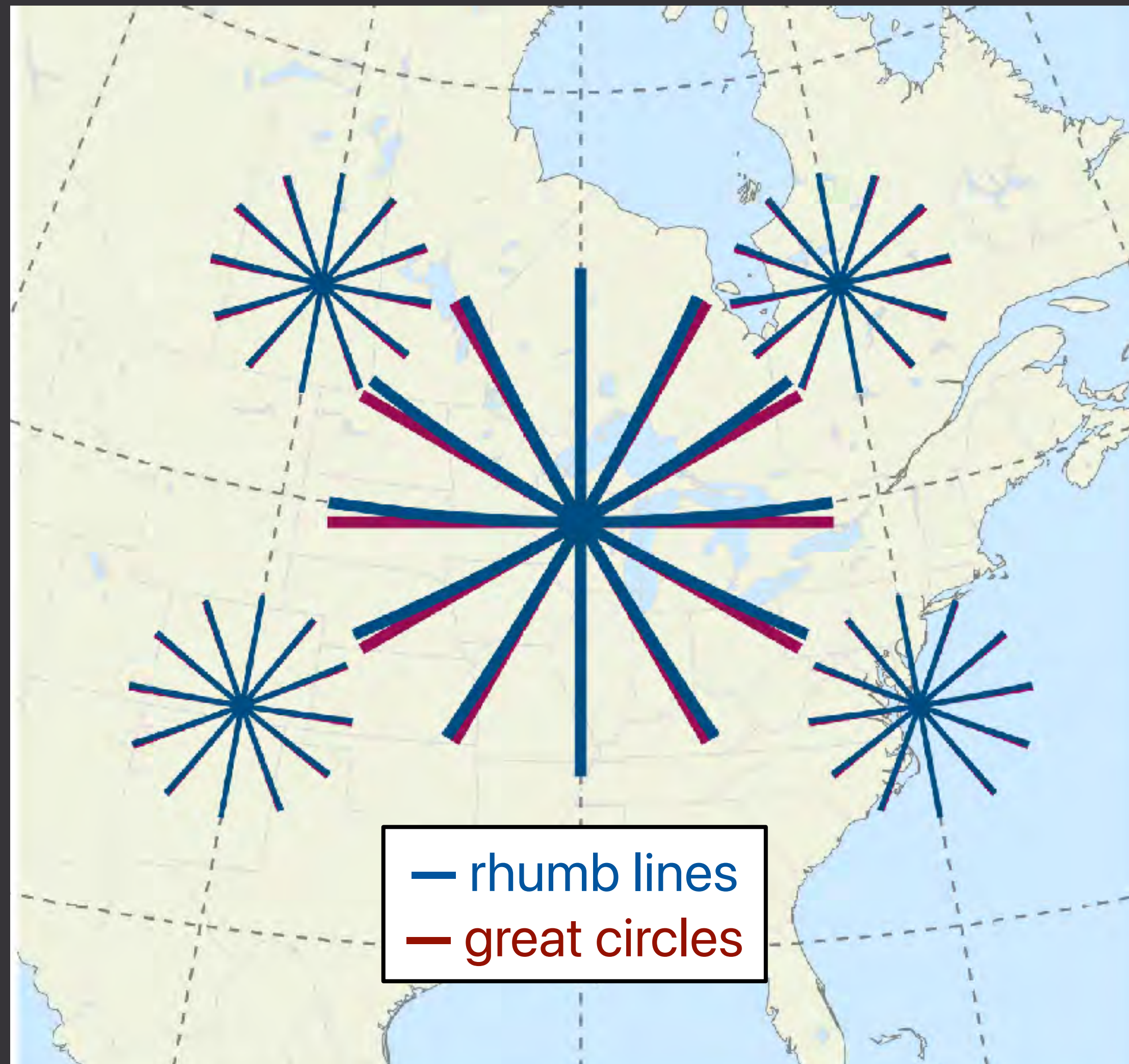


Methods: Time-distance diagram

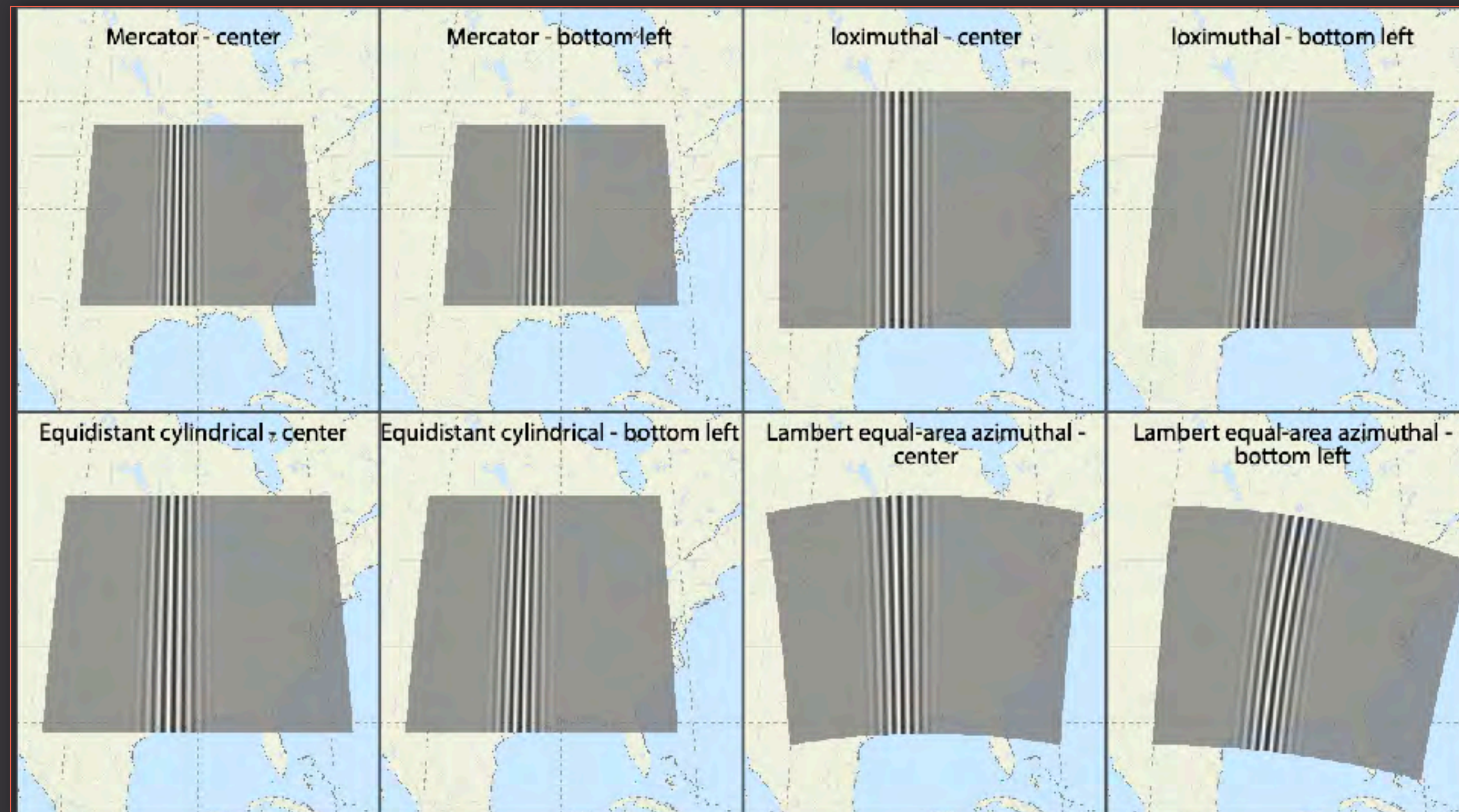
...and map projections

Gnomonic projection

- ➔ Keogram distances are measured along great circles (straight lines in the gnomonic projection)



Distances over rhumb lines and great circles on a gnomonic projection.



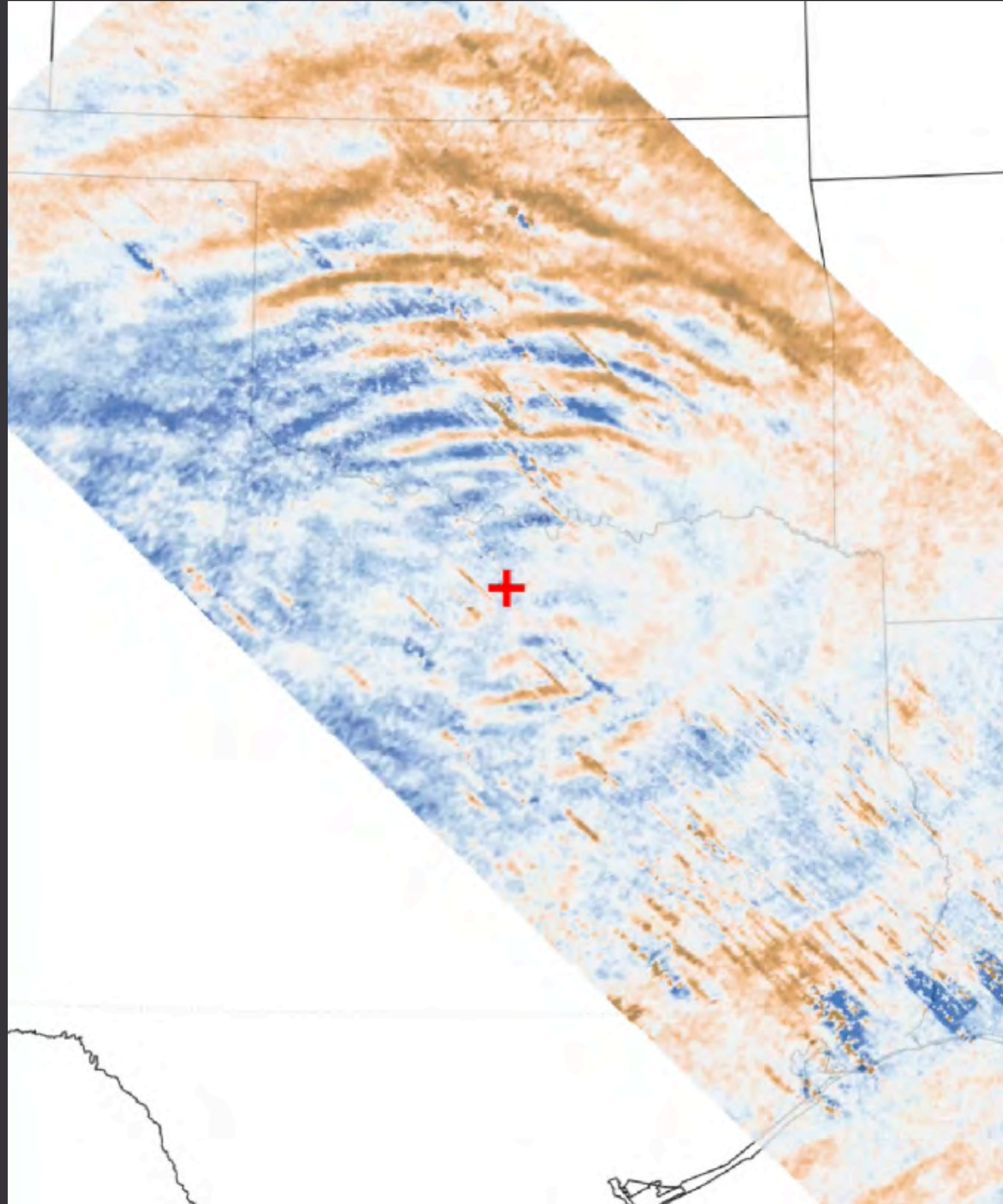
Different mapping transformations for a simple linear wave stretched along the y (North) coordinate.

The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

Methods: Time-distance diagram

...and map projections

- ➔ **Spectral analyses** and **interpolation** require an **orthonormal basis**, so everything is mapped to ECEF (Cartesian with origin at Earth's center), then onto geodetic coordinates.
- ➔ Choice of map projection is not trivial.



AWE temperature perturbations: gnomonic vs loximuthal map projections.

Methods: Time-distance diagram

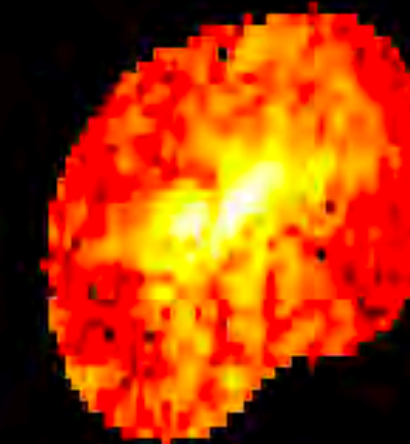
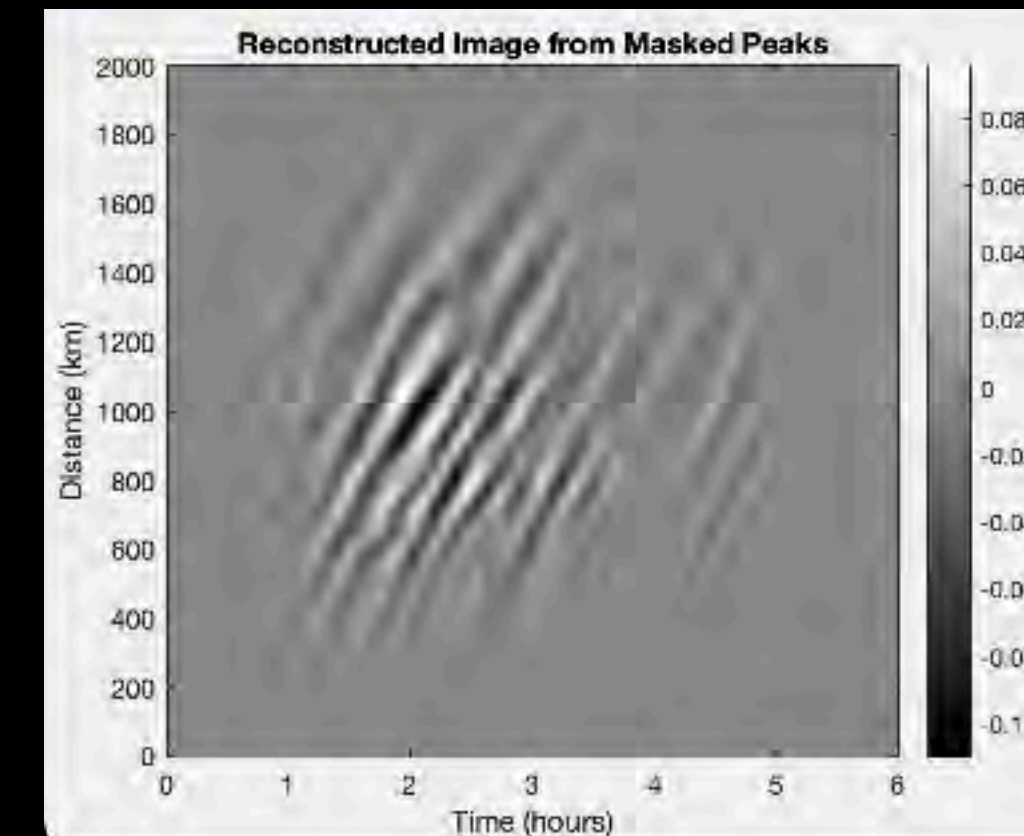
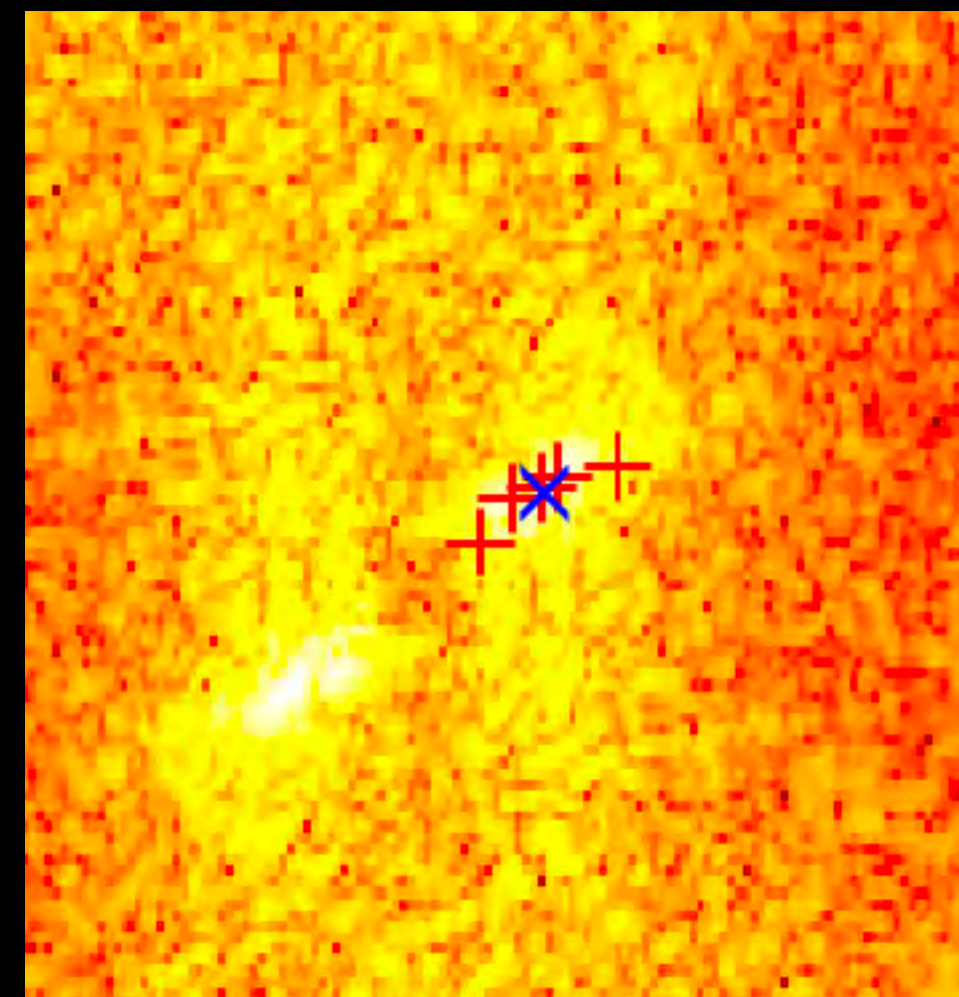
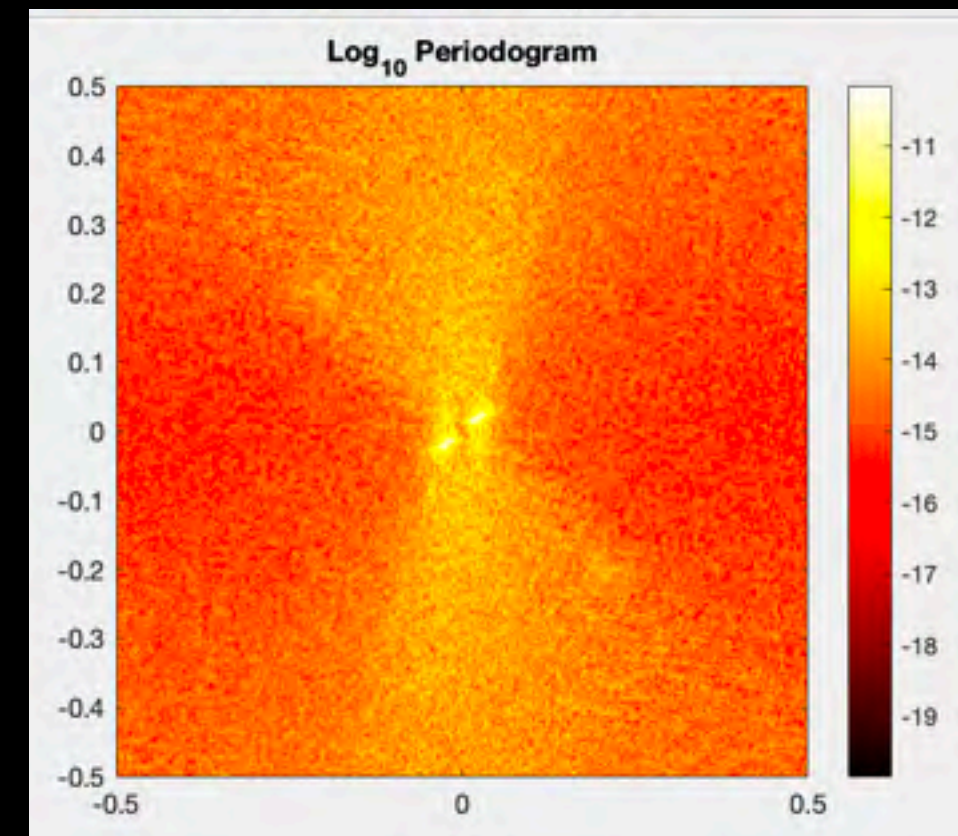
...and map projections

- ➔ **Spectral analyses** and **interpolation** require an **orthonormal basis**, so everything is mapped to ECEF (Cartesian with origin at Earth's center), then onto geodetic coordinates.
- ➔ Choice of map projection is not trivial.

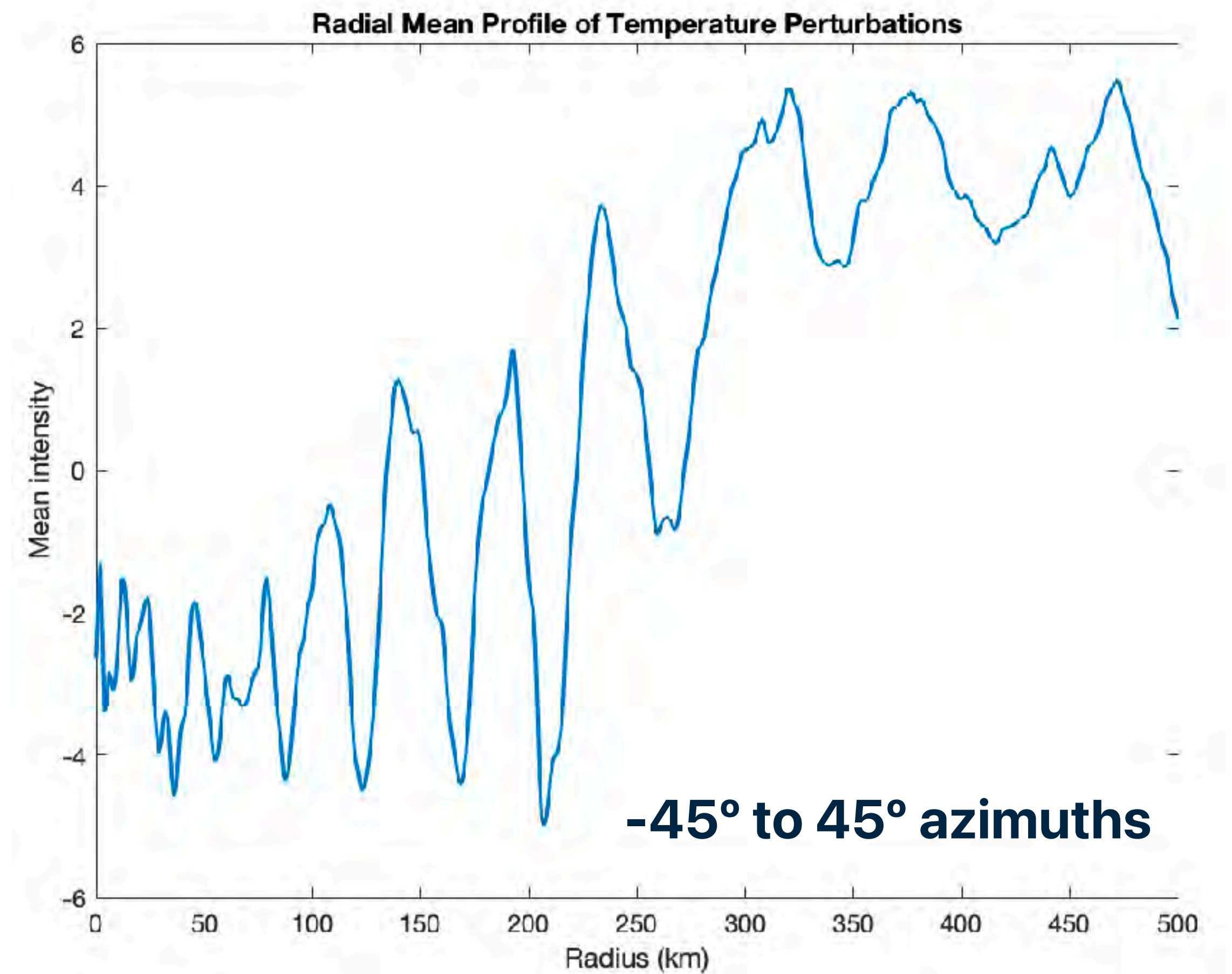
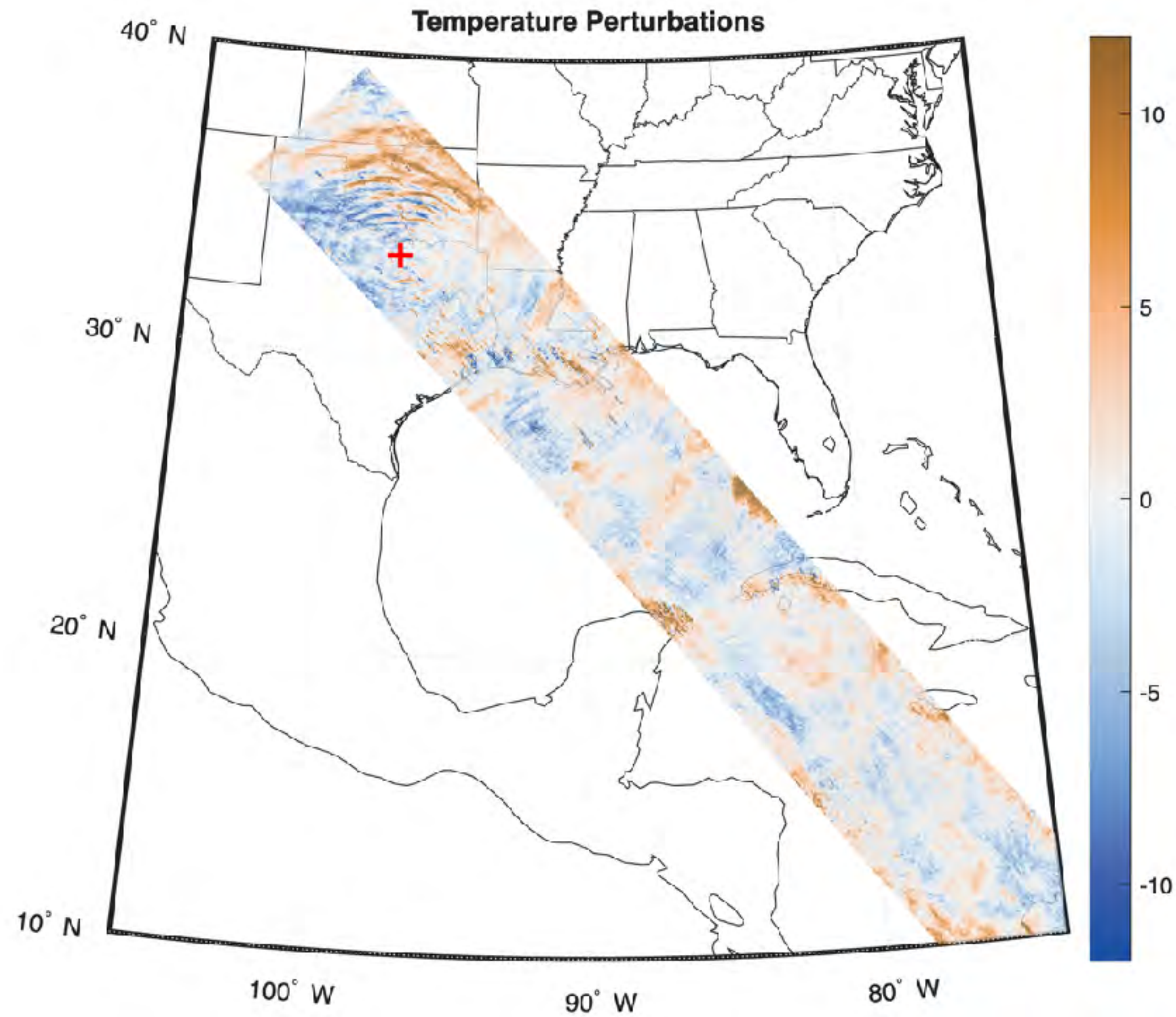
Methods: spectral analysis

★ FFT workflow:

- Periodogram.
- Peak finding.
- Energy isolation.
- Spatial reconstruction.

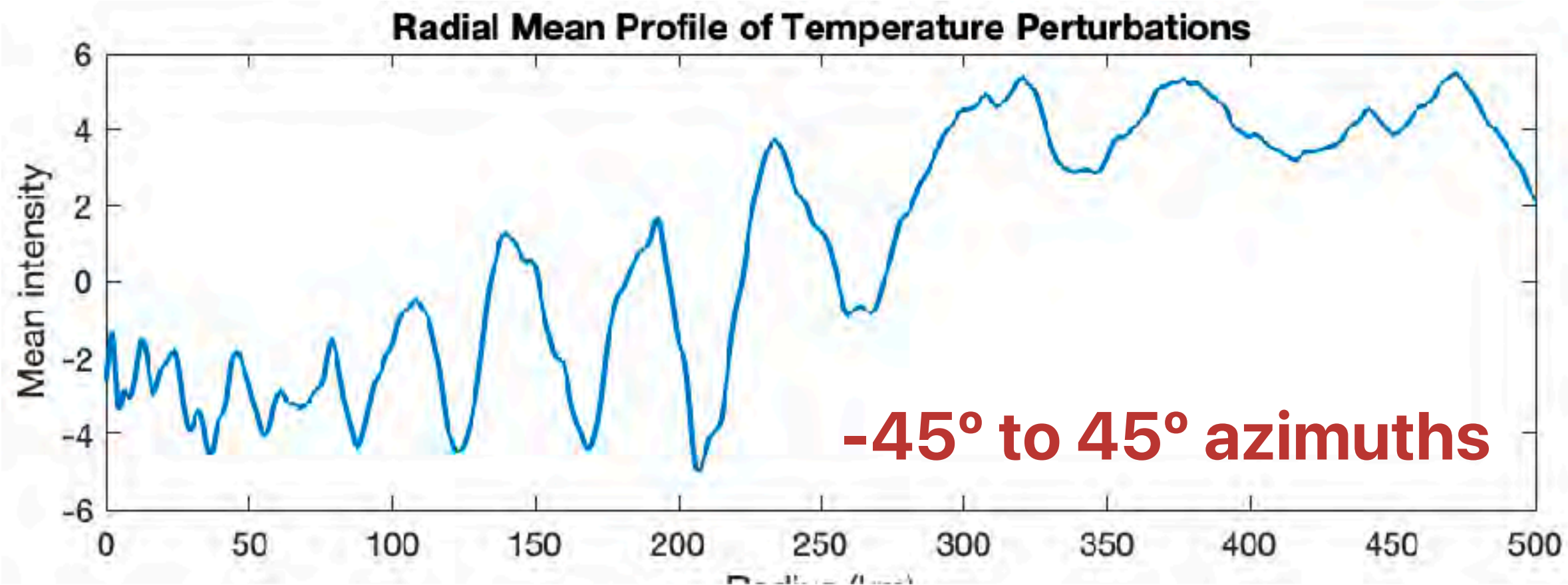
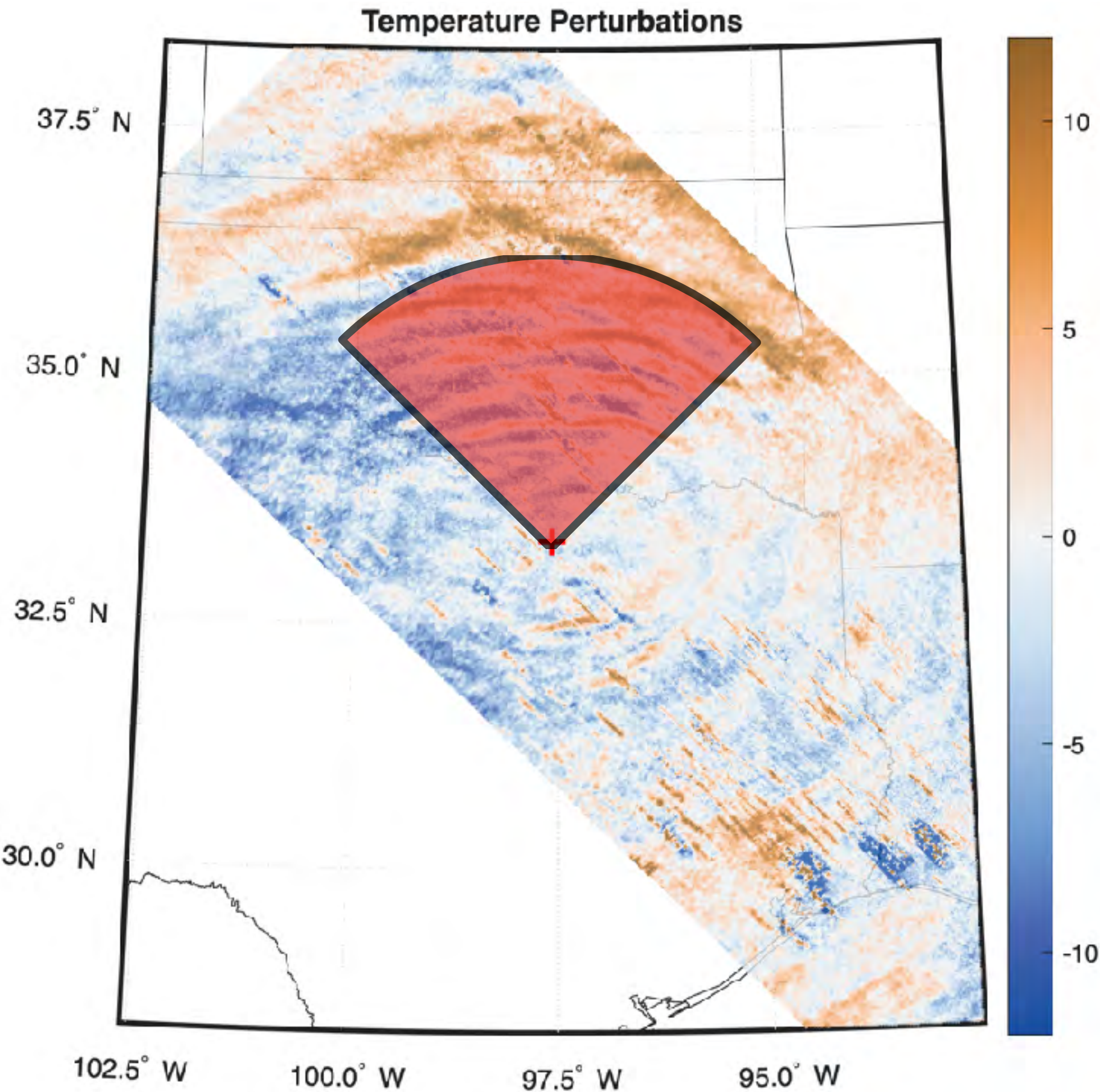


AWE **spectral analysis** with time-distance diagram from center of CGW



The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

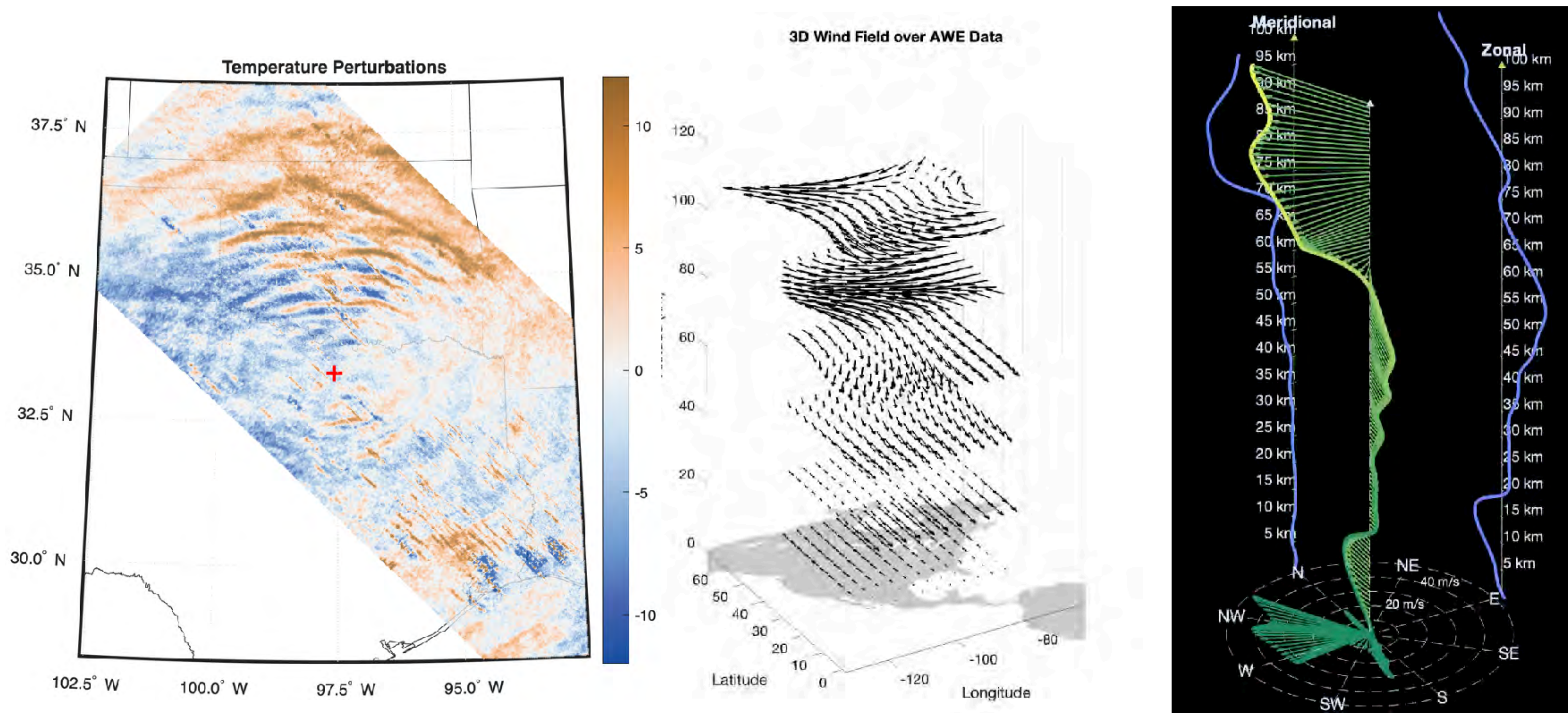
AWE spectral analysis with time-distance diagram from center of CGW



peak #	f_x (1/km)	λ (km)	amplitude ($\Delta vTEC$)	Energy (%)
1	0.0220	45.55	0.9938	14.96
2	0.0140	71.57	0.4509	9.26
3	0.0339	29.47	0.1413	0.76
4	0.0419	23.86	0.1198	0.25
5	0.0479	20.88	0.1125	0.19

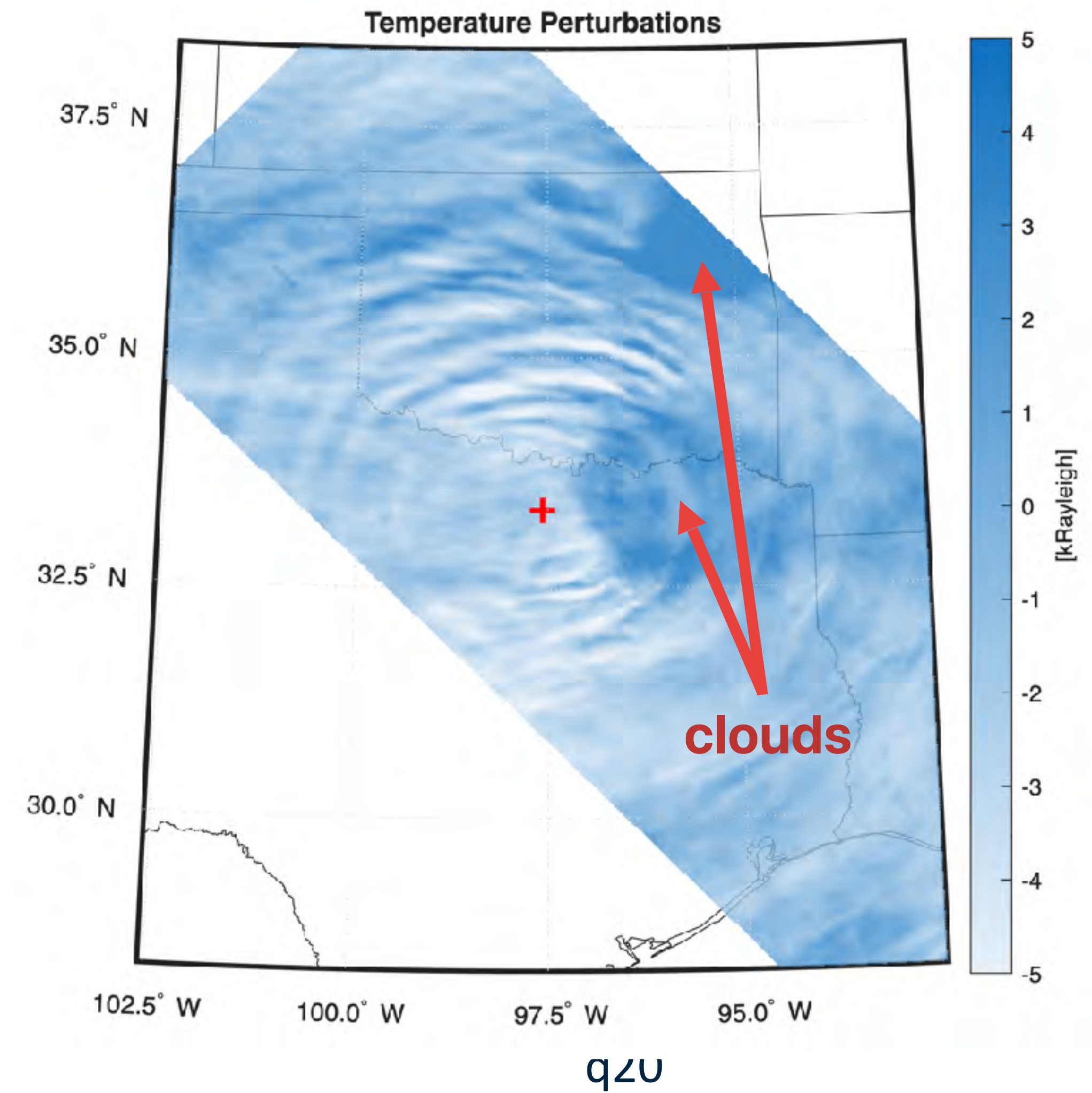
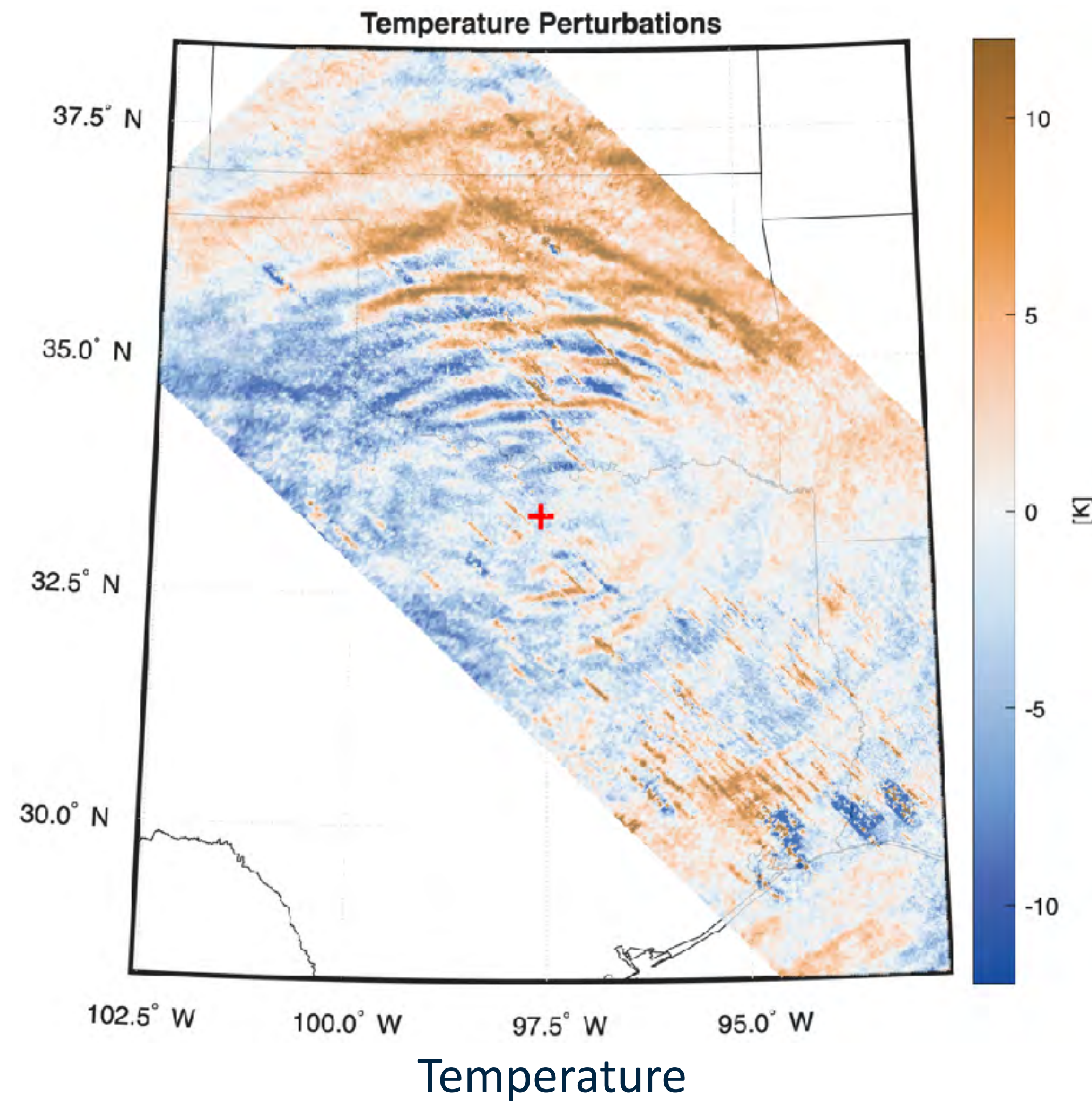
The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

AWE temperature perturbations and HWM winds



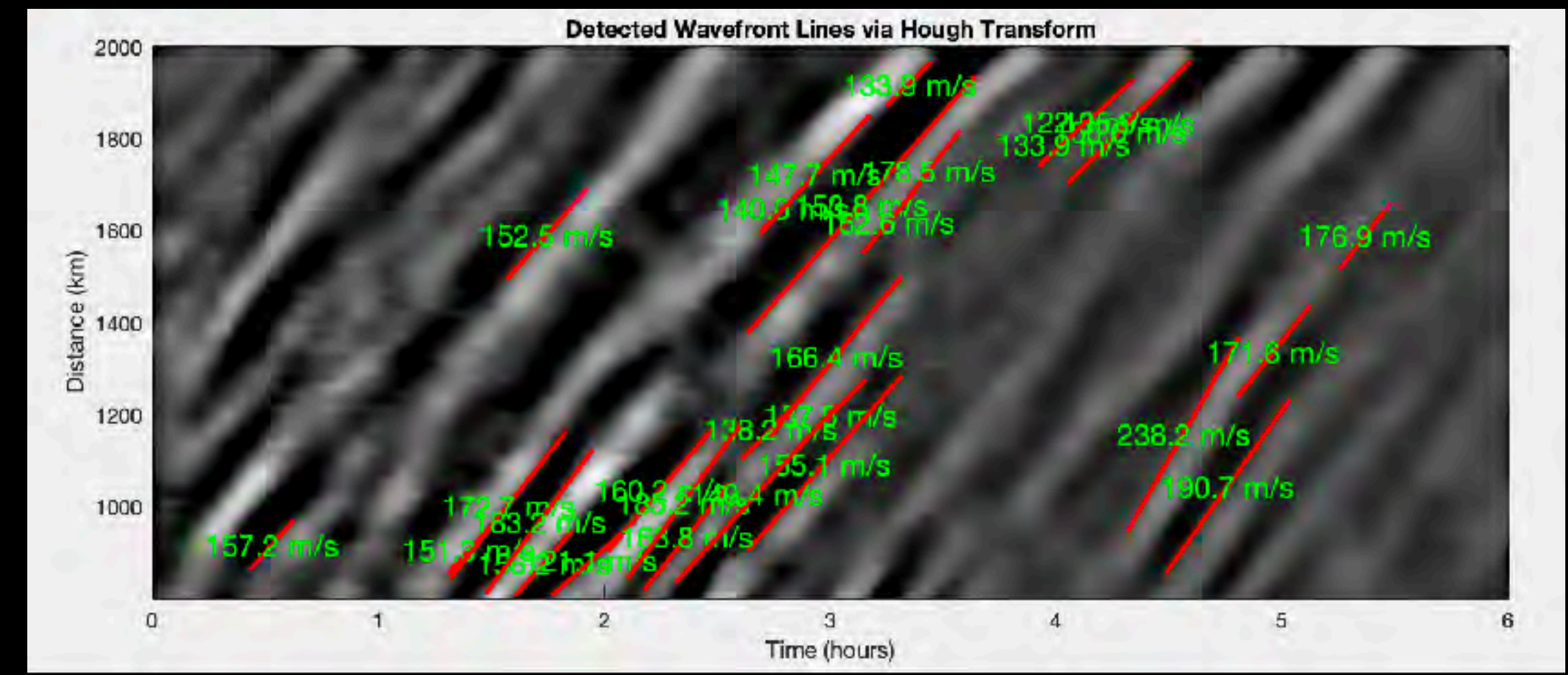
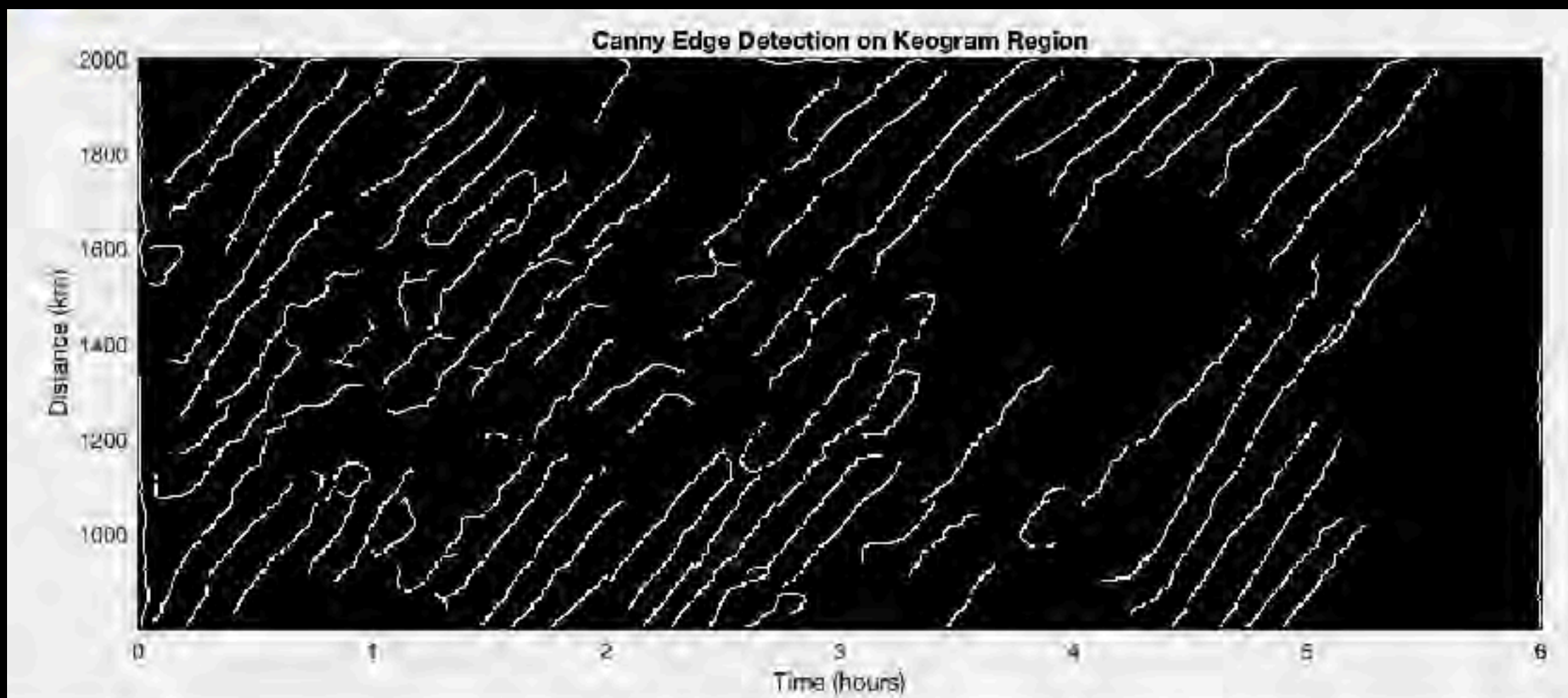
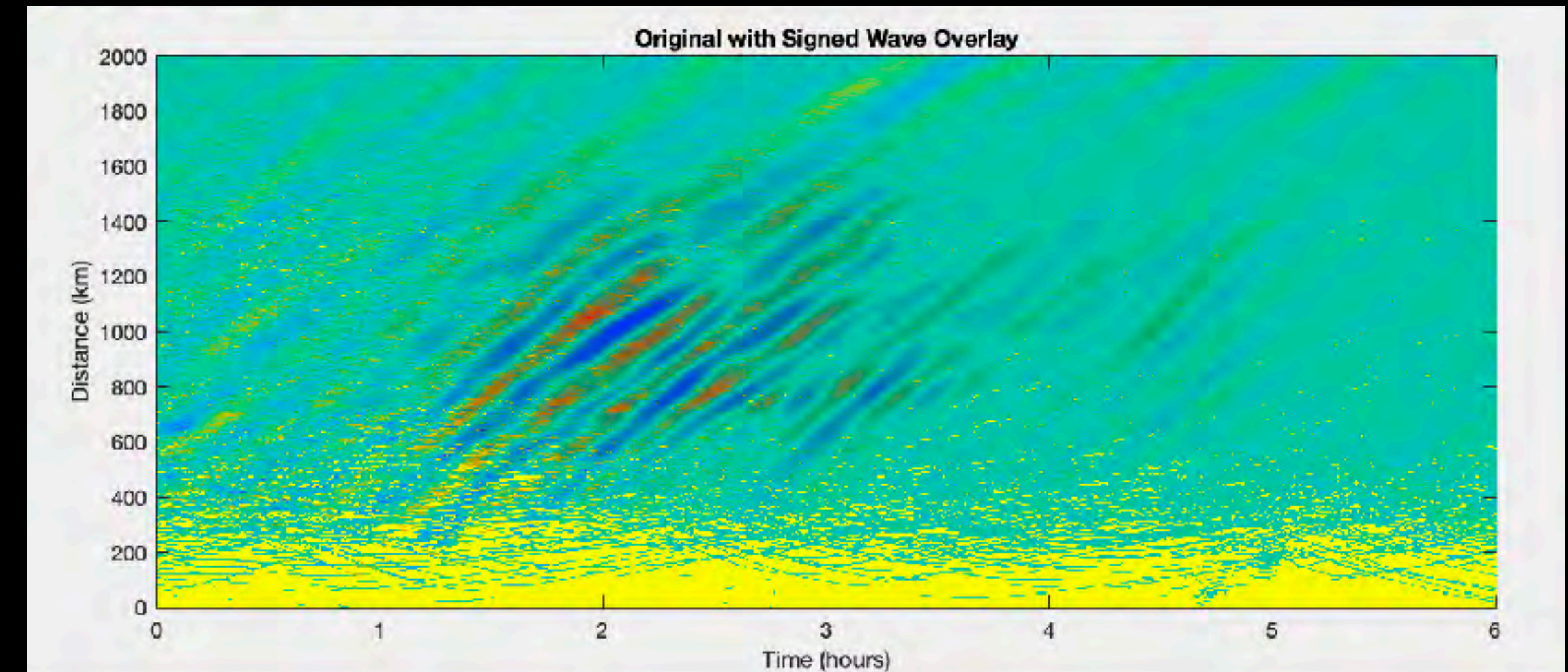
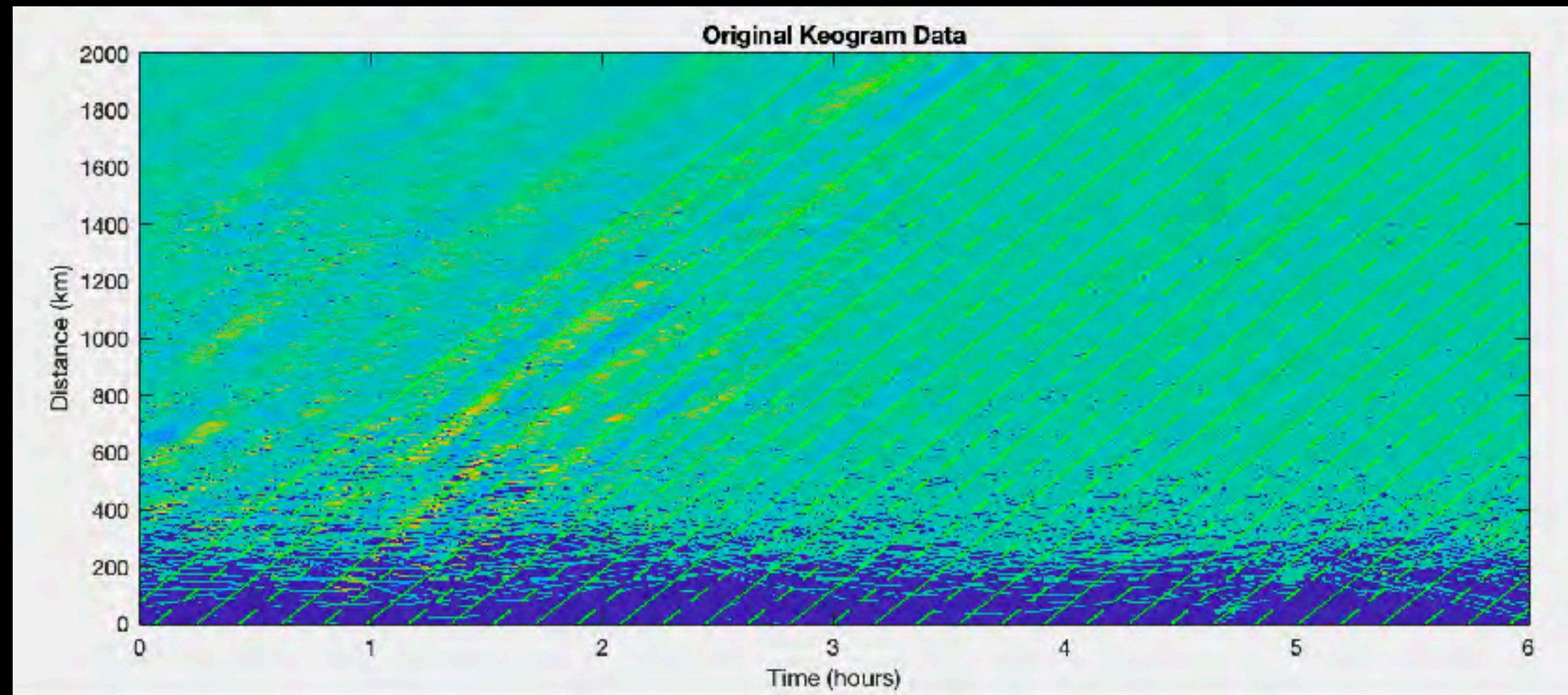
The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

AWE temperature vs radiance (q20)



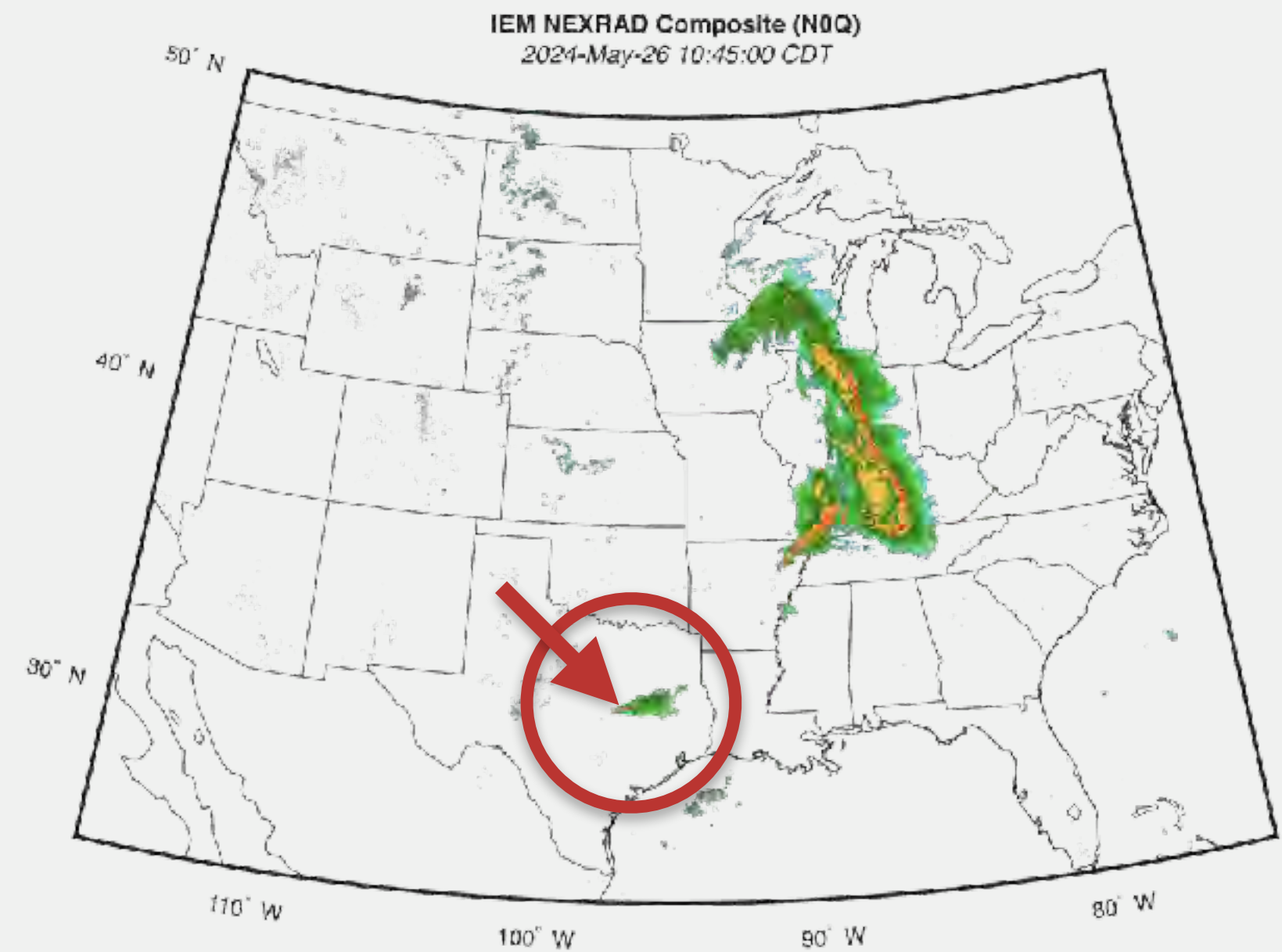
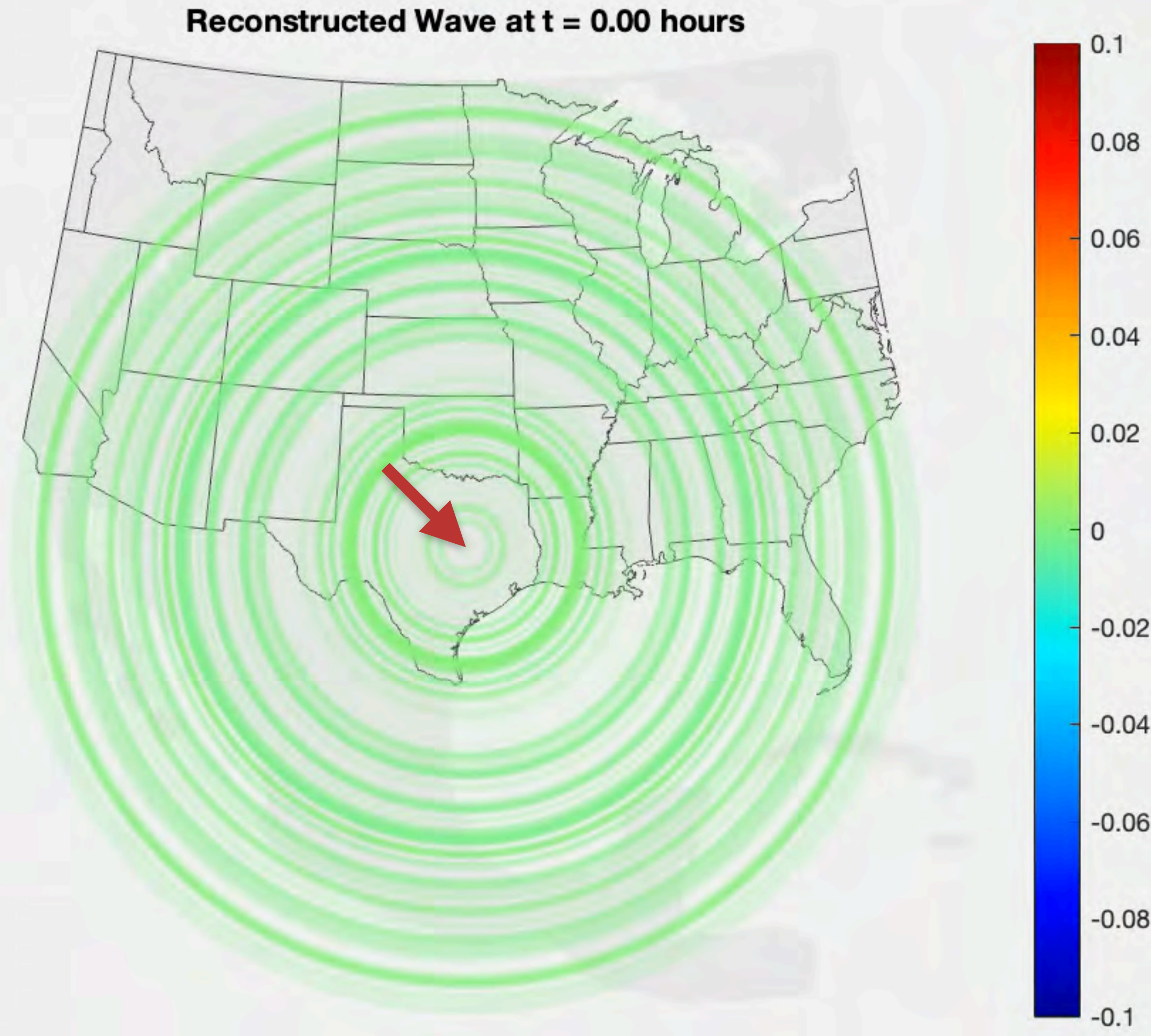
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GNSS spectral analysis + edge finding velocities



The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

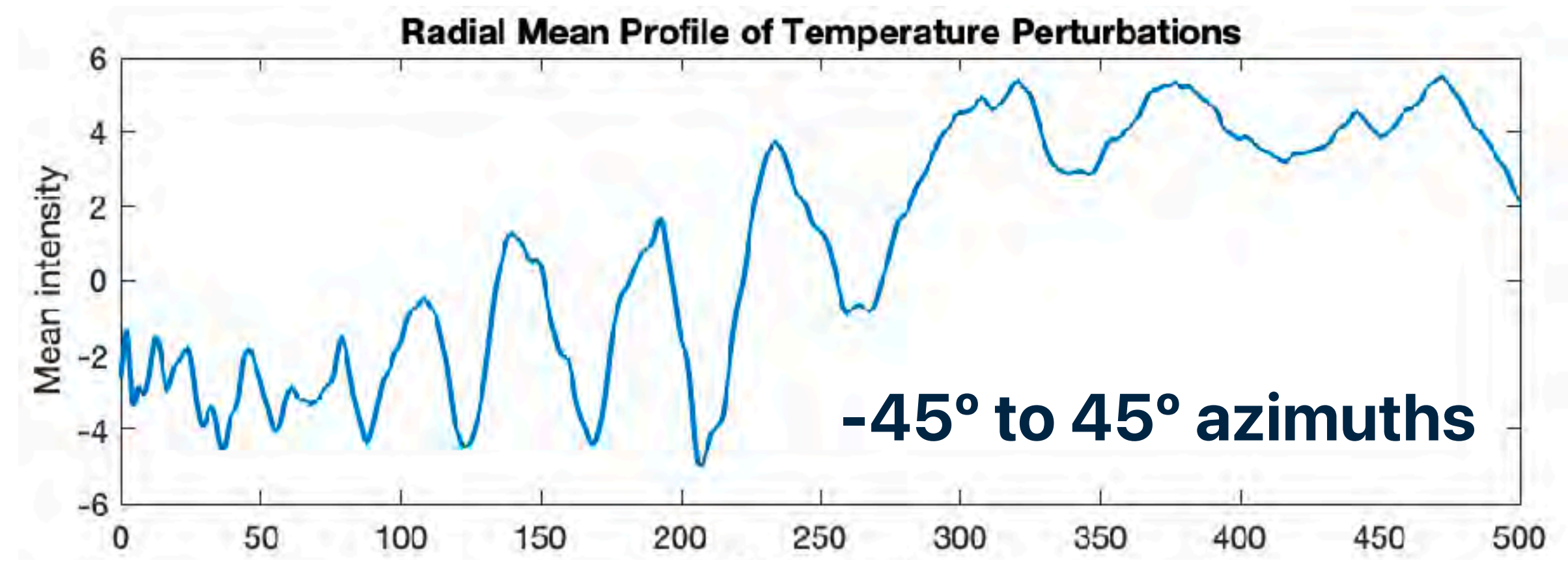
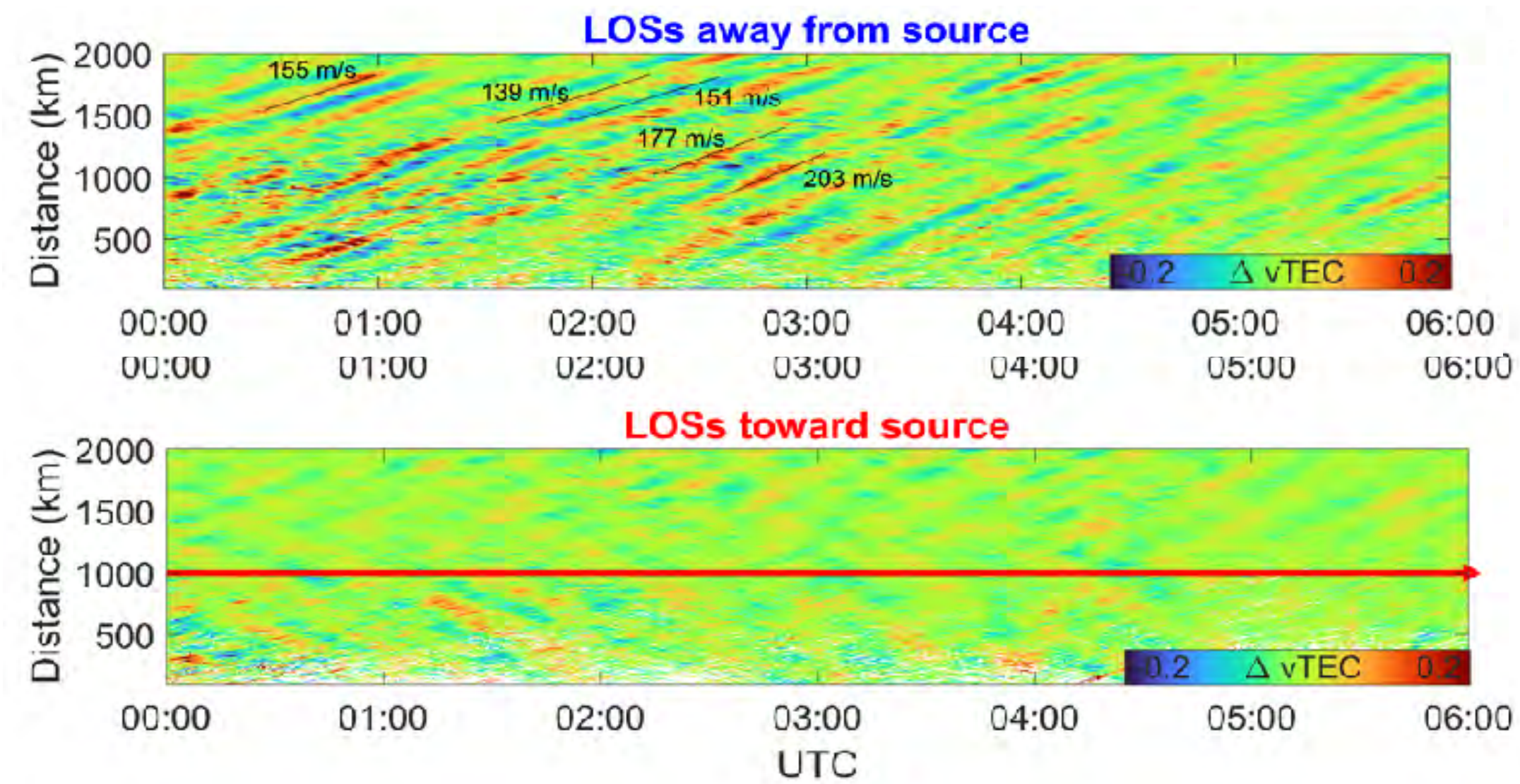
GNSS TID reconstruction @ convective cell, 15:45 UTC



Parameter	value	Error	Units
f_x	0.0169	$\pm 8.56\text{E-}04$	1/s
f_y	0.0107	$\pm 9.14\text{E-}04$	1/km
λ	375.41	± 32.09	km
T	29.66	± 1.51	min
amplitude	0.056	± 0.29	$\Delta v\text{TEC}$
phase	-25.63	-	deg
v	210.95	± 20.97	m/s

The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

GNSS vs AWE spectral analysis comparisons for May 26th, 2024 Texas cell



Parameter	value	Error	Units
f_x	0.0169	$\pm 8.56\text{E-}04$	1/s
f_y	0.0107	$\pm 9.14\text{E-}04$	1/km
λ	375.41	± 32.09	km
T	29.66	± 1.51	min
amplitude	0.056	± 0.29	$\Delta v\text{TEC}$
phase	-25.63	-	deg
v	210.95	± 20.97	m/s

peak #	f_x (1/km)	λ (km)	amplitude ($\Delta v\text{TEC}$)	Energy (%)
1	0.0220	45.55	0.9938	14.96
2	0.0140	71.57	0.4509	9.26
3	0.0339	29.47	0.1413	0.76
4	0.0419	23.86	0.1198	0.25
5	0.0479	20.88	0.1125	0.19

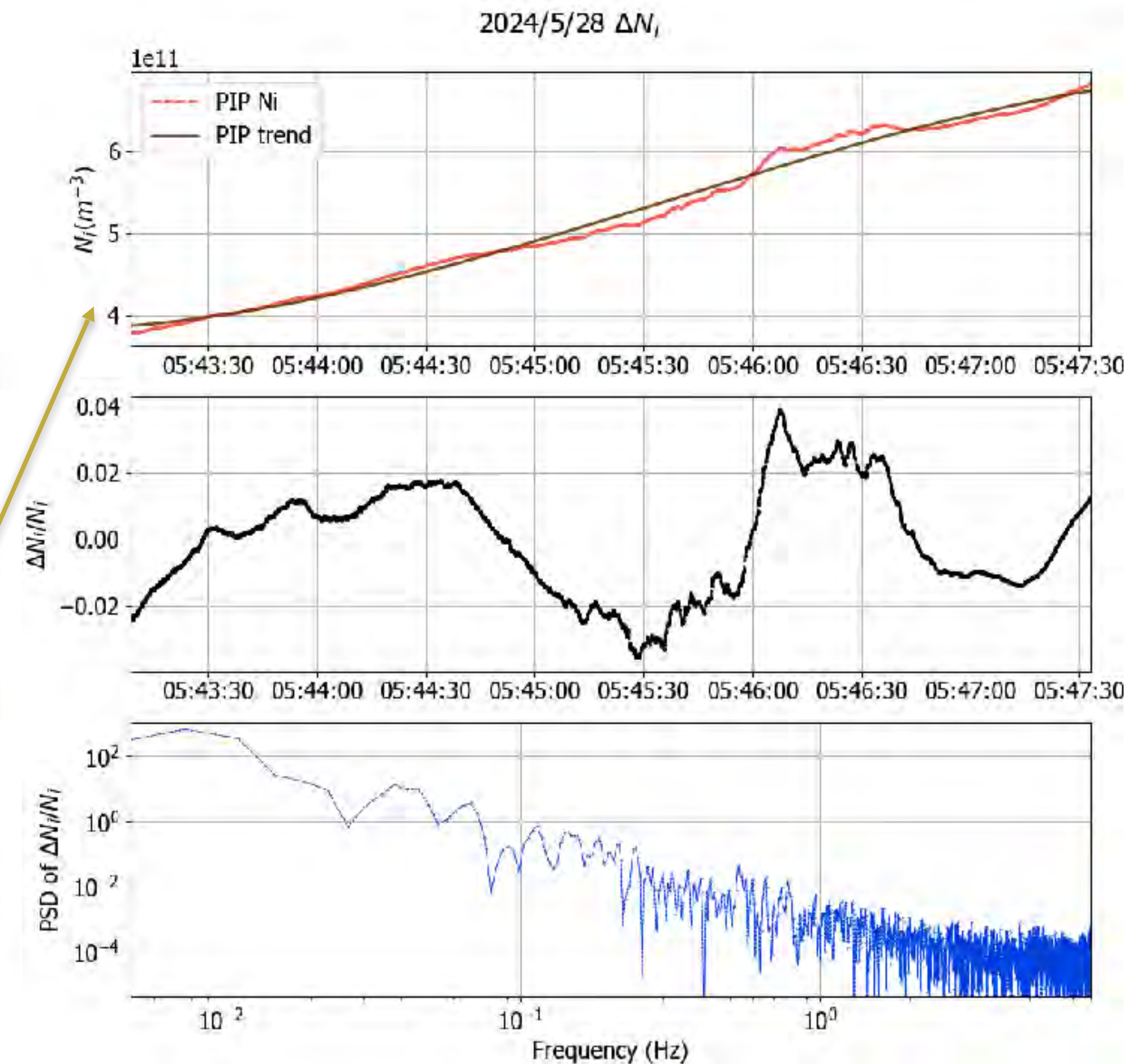
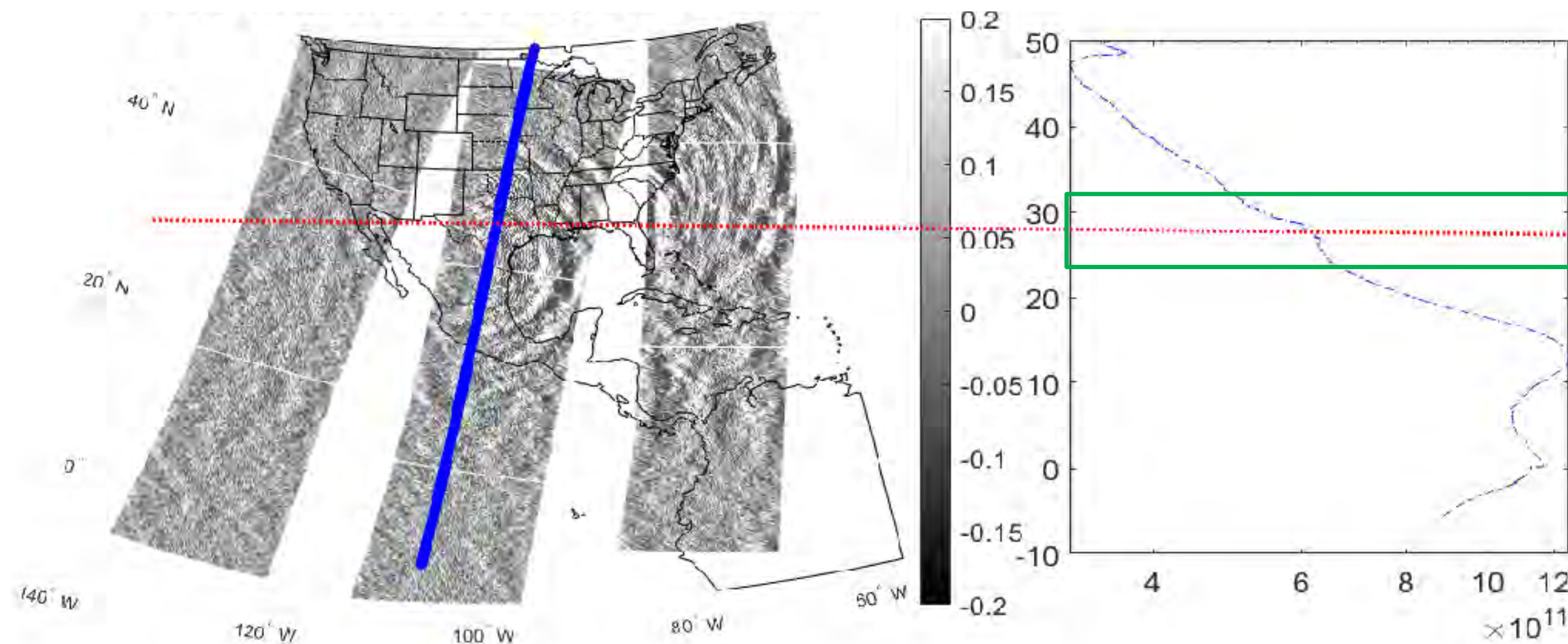
GNSS derived metrics

AWE derived metrics

The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

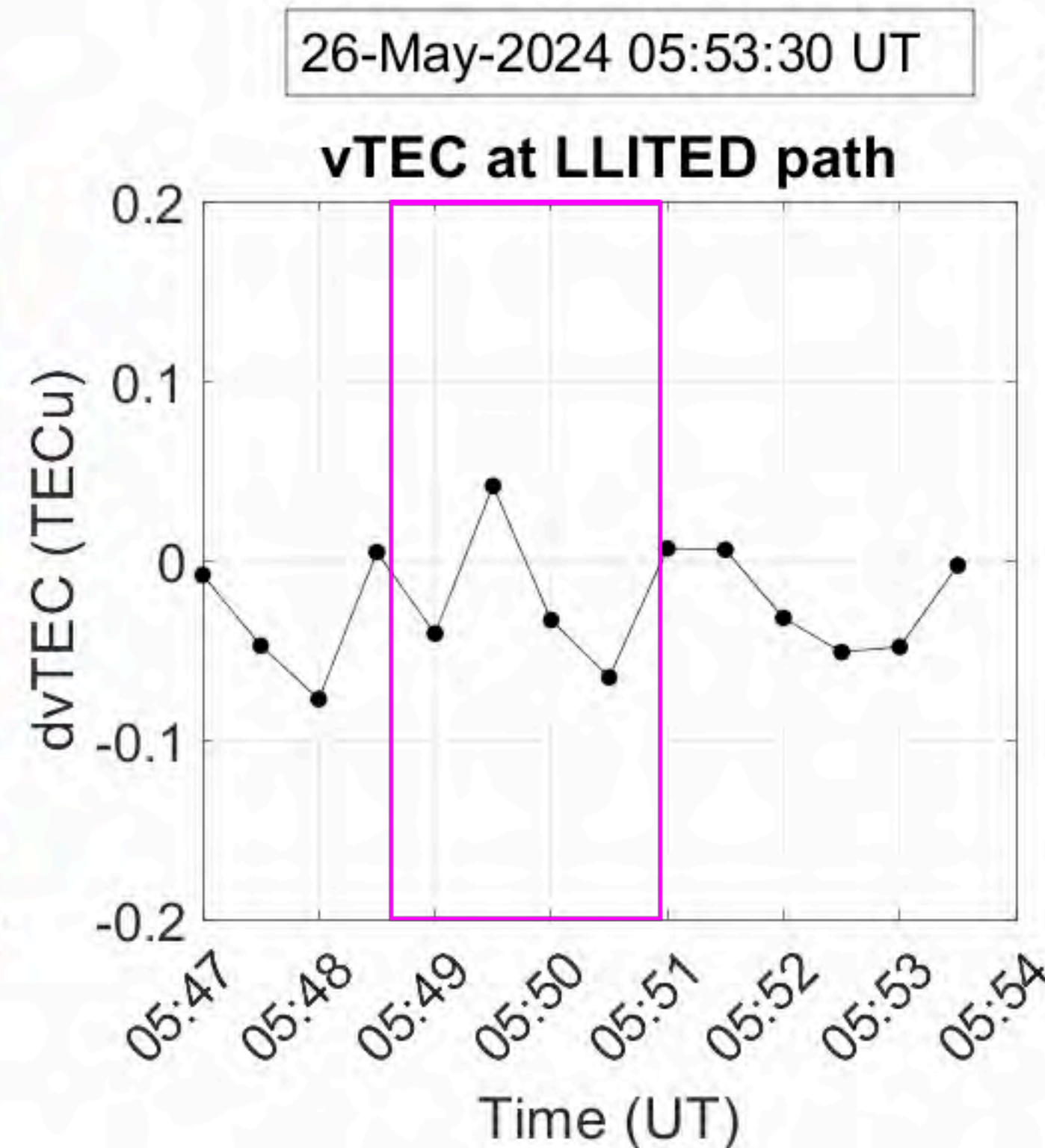
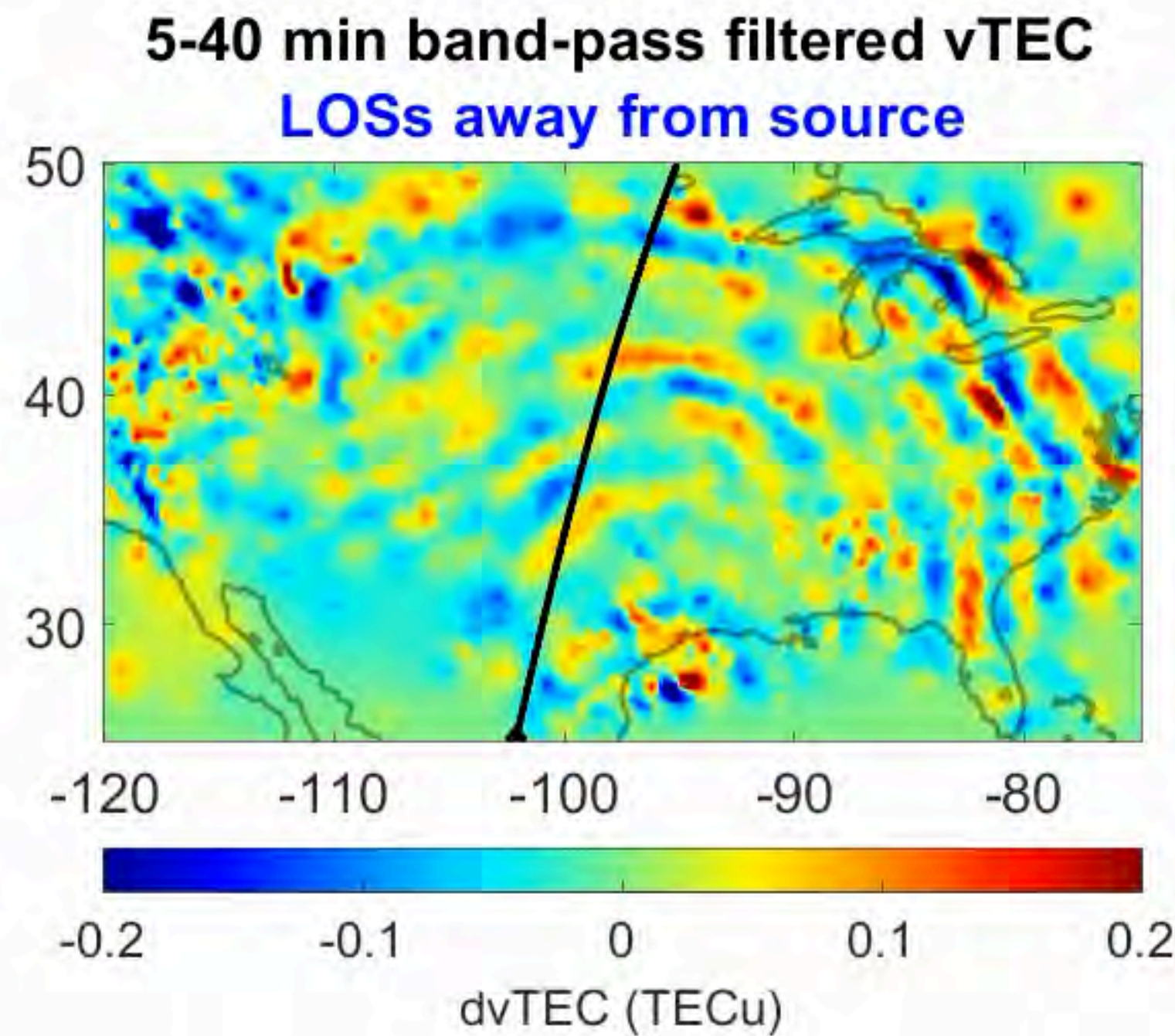
May 26 **in-situ data** in the upper atmosphere: **LLITED**

- ❑ LLITED A cubesat passed through the TIDs at an altitude of ~405km carrying planar ion probe (PIP) measuring plasma density at 16Hz
- ❑ Figure below shows the LLITED pass in blue over the AGWs observed by AIRS.
- ❑ TIDs are apparent in the in-situ data with small-scale structures along with the TIDs.

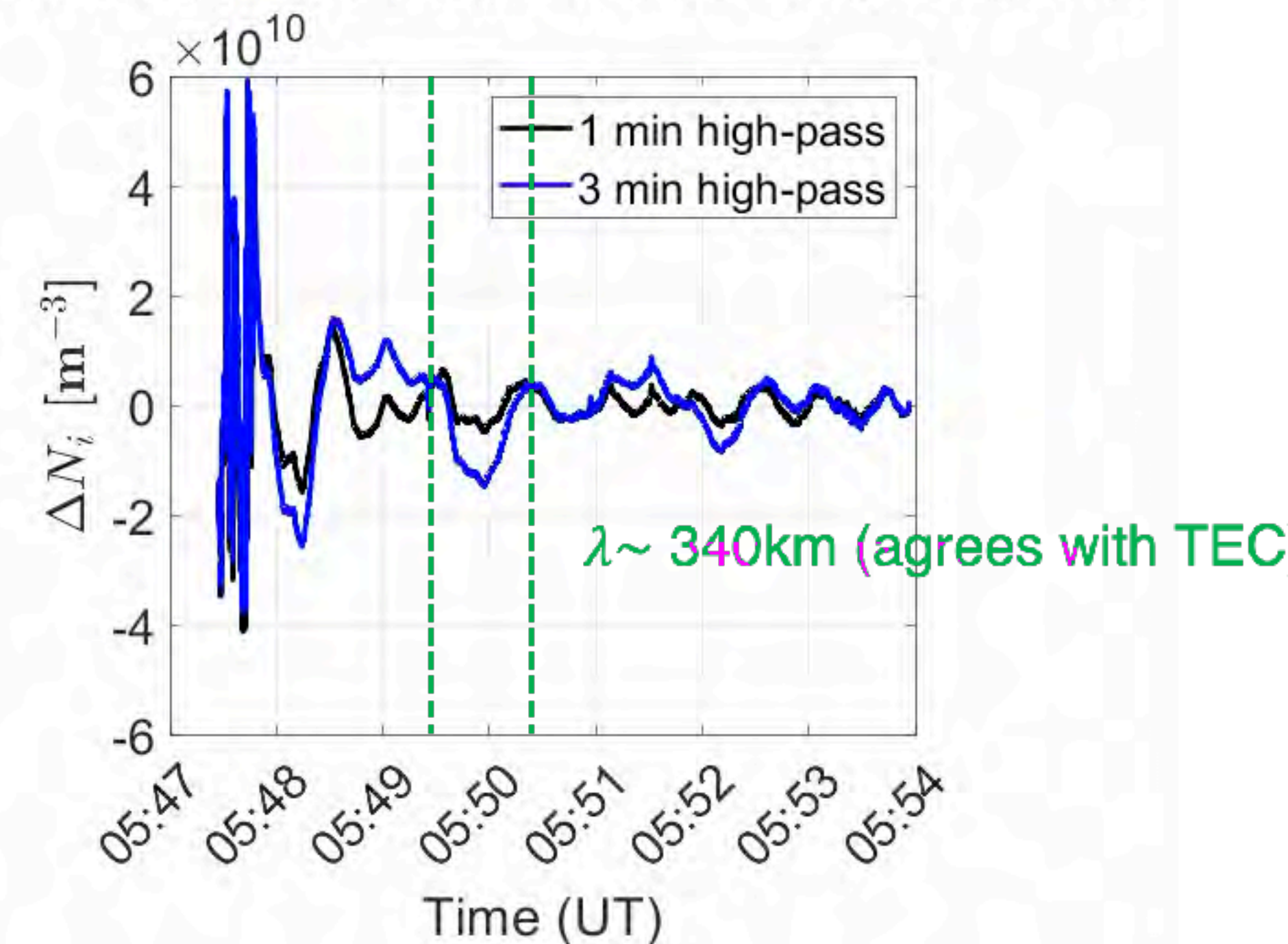


May 26 **in-situ data** in the upper atmosphere: **LLITED**

- ❑ The gravity wave generated TIDs at ~36min will be stationary to the LLITED frame.
- ❑ Thus, we can *only* compare approximate wavelengths of the TIDs from TEC and LLITED
- ❑ Detrended PIP densities show a TID with amplitude similar to MSTIDs seen in TEC which we are generated from the convective source



In-situ N_i fluctuations measured by LLITED



Products:

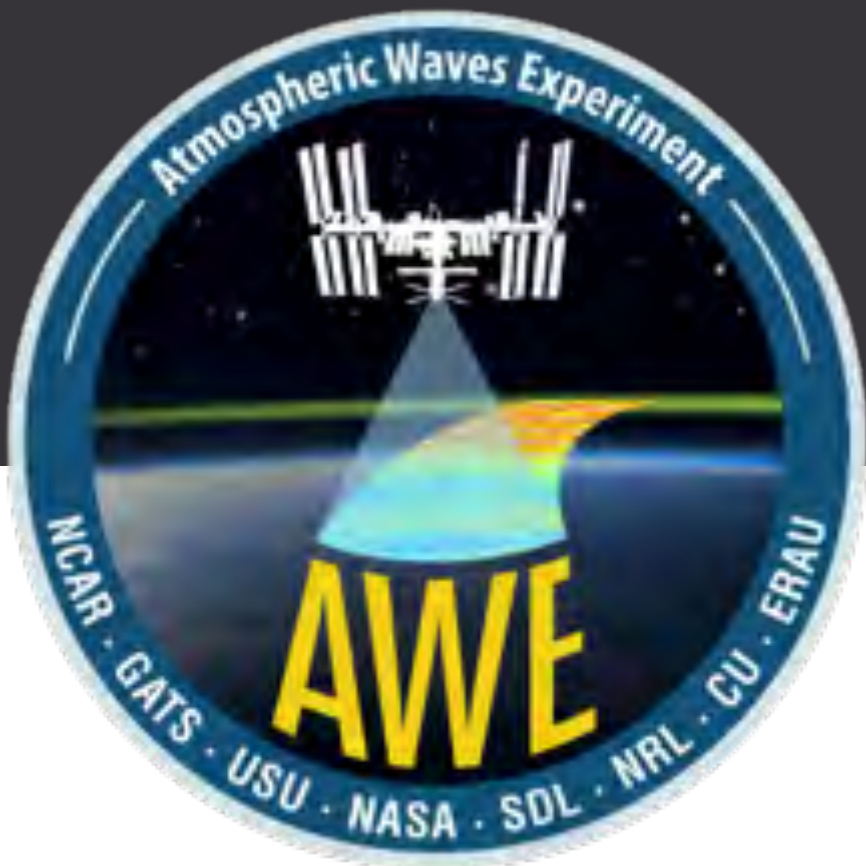
Overlay S-RAID/GNSS +
NexRad + AWE + AIRS + Modeling
& wave parameters for all data

AWE

05/26/2024 00:00:00 UT

Reconstruction of Atmospheric Gravity Waves from Deep Convection Over CONUS Using **AWE**, TEC, and AIRS Data

CEDAR 2025 June 22–27, 2025 | Des Moines, IA



Thank you!

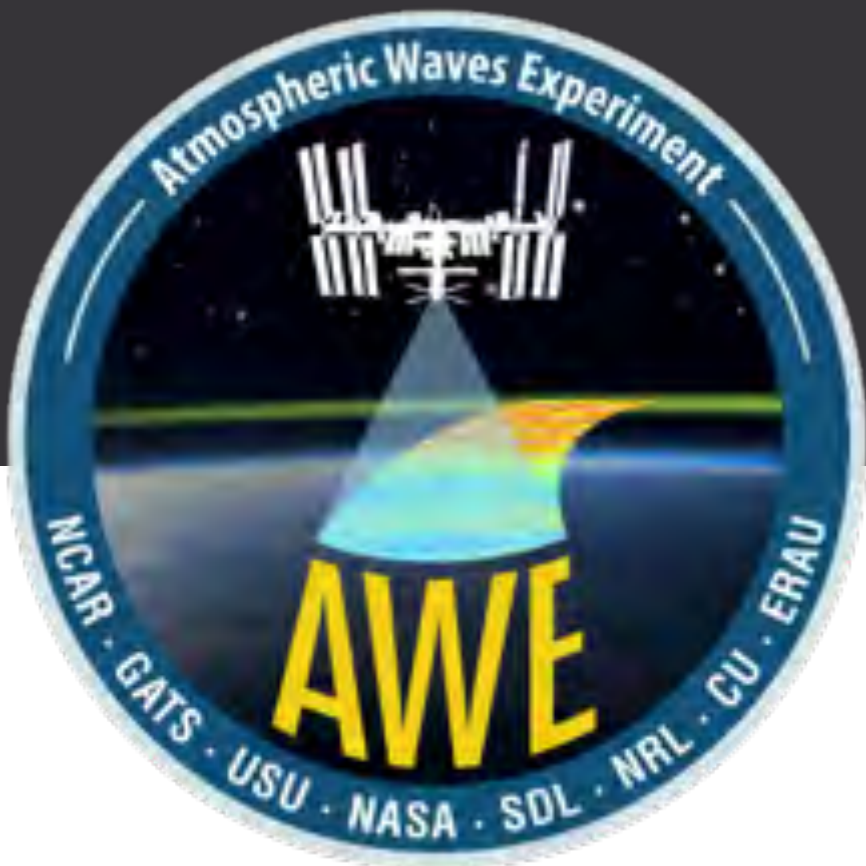
✉ *j.aguilar@erau.edu*



The material is based upon work supported by the National Aeronautics and Space Administration under Contract Number 80GSFC18C0007.

Multi-instrument Imaging of Deep Convective Events and Their Gravity Wave Responses Over CONUS Using **AWE**, AIRS, TEC, and LLITED mission

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Thank you!

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